

Original Observation Responses and the First Determinant Allocation

Complete proofs and dated programme receivers

Split-Zero research programme

19 September 2026

This successor reader gives the complete accepted calculations received and extended on 19 September 2026, with their original provider proofs and human-source citations. The accompanying three cumulative LaTeX successors retain every byte of their preceding editions by reversible insertion. This edition has no newly assigned deposited DOI.

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Part I

Original observation responses and the first determinant allocation

Complete proofs of the new source evolution, low-cutoff integrals, conductor allocation and full-window parity receivers. Earlier editions remain byte-recoverable.

Original observation response evolution

1 The unchanged packet, source and observation

Fix the original actual simple quartet and its complete arithmetic unit, period, logarithm branch and original observation. Keep $a \equiv 1 \pmod{4}$, $q = (a + 1)^2$, $c = a/2$ and the full polynomial

$$\chi(S) = \prod_{u,b=0}^a [S - c - (2u - a)\delta - i(2b - a)\gamma], \quad 0 < \delta < 1/2, \quad \gamma > 2. \quad (\text{ORE1})$$

The same proof of the finite update applies to the complete primary polynomial with multiplicities, retaining its full matrix dimension; the source-specific references below retain their own stated domains. Let $E = \mathbb{C}[S]/(\chi)$ in its original increasing-power frame. The source norm is the literal integral of $|p(c + iy)|^2$ against $m_a = w_h^{*a}(y) dy$, with $w_h = |(2\xi/h)(1/2 + iy)|^2/(2\pi)$. Let G_N be the quotient metric induced from all source polynomials of degree at most N , minimizing over all relations χh . Let $p_n(S)$ be the original monic source orthogonal polynomial, $\omega_n = \|p_n\|_{m_a}^2 > 0$, and $b_n = [p_n] \in E$. No Gamma source, rescaled mass or changed physical coordinate is substituted.

Retain the full original observation $\Lambda : E \rightarrow B = \text{im } \Lambda$, its full kernel K , and an injective frame $I : \mathbb{C}^{\dim K} \rightarrow E$. The observation includes its original period map. In particular I does not mean the native coefficient section L_I of COI11. Define, for $N \geq q - 1$,

$$\begin{aligned} H_N &= I^* G_N I, & P_N &= I H_N^{-1} I^* G_N, & Q_N &= (\Lambda G_N^{-1} \Lambda^*)^{-1}, \\ E_N &= v^* G_N v, & \theta_N &= \frac{v^* G_N P_N v}{E_N}, & v &= v_\lambda, \quad A v = \lambda v, \quad \lambda = c + a\delta + ia\gamma. \end{aligned} \quad (\text{ORE2})$$

Here A is original multiplication by S , v is the full retained class, and neither its low nor its conductor coordinate is discarded. For $K = 0$, H_N is the empty matrix, its determinant is one and $P_N = 0$. For $K = E$, $B = 0$ and the corresponding Q_N is empty. These conventions give the same formulas below. The exact minimum-lift map is $G_N^{-1} \Lambda^* Q_N : B \rightarrow E$, so

$$G_N(I - P_N) = \Lambda^* Q_N \Lambda, \quad E_N(1 - \theta_N) = (\Lambda v)^* Q_N (\Lambda v). \quad (\text{ORE3})$$

This is the equality-constrained quadratic minimum described by Boyd and Vandenberghe [1, Appendix C.4.1, p. 673]; the following calculation supplies its complex Hermitian dictionary for the actual G_N and Λ . Indeed the lift has observation equal to the prescribed value, is orthogonal to K , and hence is its unique minimum. This proves both identities on the original space, including its full cross metric.

2 A single actual source increment and both determinant allocations

At the following cutoff put $b = b_{N+1}$, $\omega = \omega_{N+1}$. The exact source orthogonal decomposition $\mathcal{P}_{N+1} = \mathcal{P}_N \perp \mathbb{C}p_{N+1}$ implies

$$G_{N+1}^{-1} = G_N^{-1} + \frac{bb^*}{\omega}, \quad G_{N+1} = G_N - \frac{G_N bb^* G_N}{\omega + \beta_N}, \quad \beta_N = b^* G_N b. \quad (\text{ORE4})$$

For the first equality, the squared dual norm of a functional on E is the sum of its squared values on a source orthogonal basis divided by the original ω_n ; the new term is exactly bb^*/ω . For the second, multiply the proposed right side by the first matrix. The coefficient of $G_N bb^*/\omega$ cancels, proving the inverse without any assumed metric equivalence. This is the exact rank-one inverse identity recorded by Hager [2, p. 221, equation (2)], with its source and denominator derived here.

On the original retained eigenclass domain NCE9 proves $b^* G_N v \neq 0$, so $\beta_N > 0$. Define

$$r_{N+1} = \frac{\omega}{\omega + \beta_N} = \frac{\det G_{N+1}}{\det G_N}, \quad \xi_N = 1 - r_{N+1} \in (0, 1),$$

$$\alpha_N = \frac{b^* G_N P_N b}{\beta_N} \in [0, 1], \quad d_{K,N} = \frac{\det H_{N+1}}{\det H_N}. \quad (\text{ORE5})$$

The determinant lemma follows by taking a basis whose first vector is the single nonzero column of the update, or by expansion of the determinant in that rank-one column. Applied to H_N and (ORE4), it gives

$$\boxed{d_{K,N} = 1 - \xi_N \alpha_N, \quad \frac{\det Q_{N+1}}{\det Q_N} = \frac{r_{N+1}}{d_{K,N}}.} \quad (\text{ORE6})$$

For the second equation use (ORE3) with b : $(\Lambda b)^* Q_N \Lambda b = \beta_N (1 - \alpha_N)$. Updating Q_N^{-1} by $(\Lambda b)(\Lambda b)^*/\omega$ then proves the equation with denominator $\omega + \beta_N (1 - \alpha_N) = (\omega + \beta_N) d_{K,N}$. Thus $r_{N+1} \leq d_{K,N} \leq 1$ and both determinant ratios are positive, even when one kernel or image is zero dimensional. The exact original per-step logarithmic allocations are

$$-\log d_{K,N}, \quad -\log r_{N+1} + \log d_{K,N}. \quad (\text{ORE7})$$

Their sum is $-\log r_{N+1}$. These formulas compute the restriction and quotient of the same original metric; they do not replace the original directional α_N by a rank fraction.

3 The two boundary responses form one constrained sequence

Use the original definitions, with the same full v ,

$$U_N = \frac{b_N^* G_N P_N v}{b_N^* G_N v}, \quad V_N = \frac{b_{N+1}^* G_N P_N v}{b_{N+1}^* G_N v}. \quad (\text{ORE8})$$

These are exactly NCE14 and EC37; neither row has been exchanged. Put $s = b^* G_N v$ and $s_B = b^* G_N (I - P_N) v = (1 - V_N) s$. The update (ORE4) gives $b^* G_{N+1} v = r_{N+1} s$. The attained quotient inverse update gives

$$\begin{aligned} Q_{N+1} &= Q_N - \frac{Q_N \Lambda b b^* \Lambda^* Q_N}{\omega + \beta_N (1 - \alpha_N)}, \\ b^* G_{N+1} (I - P_{N+1}) v &= \frac{r_{N+1}}{d_{K,N}} s_B. \end{aligned} \quad (\text{ORE9})$$

At cutoff $N + 1$ its first boundary class is literally the same b . Divide the second equality by $r_{N+1} s \neq 0$ to obtain

$$\boxed{U_{N+1} = 1 - \frac{1 - V_N}{d_{K,N}}.} \quad (\text{ORE10})$$

This is an exact original-source relation between the two response rows at adjacent cutoffs. It includes all conductor terms through the actual H_N and every P_N . No real part, complex conjugation, phase or normalization is introduced in this transport.

In particular let $R_N = 1 - V_N$ and, for $N \geq q$, retain the three original products at cutoff N :

$$\begin{aligned} P_{B,N} &= \frac{\overline{R_{N-1}} R_N}{d_{K,N-1}}, \\ P_{K,N} &= \left(1 - \frac{\overline{R_{N-1}}}{d_{K,N-1}}\right) (1 - R_N), \quad P_{\times,N} = 1 - P_{K,N} - P_{B,N}. \end{aligned} \quad (\text{ORE11})$$

The first is $(1 - \overline{U}_N)(1 - V_N)$ by (ORE10); the second is $\overline{U}_N V_N$. Therefore the actual centered pairings are

$$\Phi_{X,N} = 2E_N \Re(\mathbf{c}_{N,\lambda} P_{X,N}), \quad \sum_X \Phi_{X,N} = 2a\delta E_N, \quad X \in \{K, B, \times\}. \quad (\text{ORE12})$$

This is the NCE15 current, now with its consecutive response factors and determinant denominator explicitly linked. The shared phase error also keeps its signed sum zero by $\sum_X P_{X,N} = 1$. At the first cutoff U_{q-1} remains its original first boundary value; no fictitious previous quotient of dimension q is inserted.

4 Exact class energy and observation energy evolution

Define the original second boundary energy and its actual class loss

$$p_{2,N} = \frac{|b^* G_N v|^2}{E_N \beta_N}, \quad \eta_N = \xi_N p_{2,N}, \quad h_N = \frac{E_{N+1}}{E_N} = 1 - \eta_N. \quad (\text{ORE13})$$

The last equality is direct substitution in (ORE4). Positivity of G_{N+1} gives $h_N > 0$. In fact $0 < p_{2,N} \leq 1$ and $0 < \eta_N < 1$. The Cauchy–Schwarz upper bound follows by minimizing the full positive square of $v - tb$; it uses the original G_N . Applying (ORE9) to the fixed observation value Λv proves

$$(1 - \eta_N)(1 - \theta_{N+1}) = 1 - \theta_N - \frac{\eta_N |1 - V_N|^2}{d_{K,N}}. \quad (\text{ORE14})$$

Indeed the quotient energy drop is precisely $|s_B|^2/[\omega + \beta_N(1 - \alpha_N)]$; divide by E_N , use (ORE13) and retain its denominator from (ORE6). Rearranging gives the same exact update for the kernel component:

$$\theta_{N+1} - \theta_N = \frac{\eta_N}{1 - \eta_N} \left[\theta_N - 1 + \frac{|1 - V_N|^2}{d_{K,N}} \right]. \quad (\text{ORE15})$$

Thus a small complete-class increment alone does not control the change of the original kernel projection: its actual amplification and its restriction determinant appear in the exact same equation. The nonnegative quotient energy gives the finite evaluated bound

$$0 \leq \frac{\eta_N |1 - V_N|^2}{d_{K,N}} \leq 1 - \theta_N. \quad (\text{ORE16})$$

If $\Lambda v = 0$, both sides of the underlying energy drop vanish, $\theta_N = 1$ and $U_N = V_N = 1$ for every cutoff. If $\Lambda v \neq 0$, the quotient energy is strictly positive at every cutoff by $Q_N > 0$. This resolves the possible zero value without deleting that original class or taking an undefined logarithm.

There is also an exact propagation of the two original boundary energies. For the unchanged $b = b_{N+1}$, (ORE4) gives $b^* G_{N+1} b = r_{N+1} \beta_N$ and $b^* G_{N+1} v = r_{N+1} s$. Hence

$$p_{1,N+1} = \frac{r_{N+1} p_{2,N}}{1 - (1 - r_{N+1}) p_{2,N}}. \quad (\text{ORE17})$$

Combining this with the independent original parity calculation in the companion boundary-energy proof evaluates both energies from the same consecutive source moments, with no metric switch.

5 Both windows and the full original coefficient transition

For any positive sequence f_N , direct cancellation of consecutive ratios proves

$$\log \frac{f_{2q-1} f_{2q}}{f_{q-1} f_q} = \sum_{N=q-1}^{2q-1} w_N \log \frac{f_{N+1}}{f_N}, \quad w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \text{ otherwise.} \quad (\text{ORE18})$$

Apply it to $\det G_N$, $\det H_N$, $\det Q_N$ and E_N . Their increments are respectively $\log r_{N+1}$, $\log d_{K,N}$, $\log r_{N+1} - \log d_{K,N}$ and $\log(1 - \eta_N)$. Both original windows and each determinant frame cancellation are thereby retained. If $\Lambda v \neq 0$, it also applies to $E_N(1 - \theta_N)$ with its exact ratio (ORE14). If $\Lambda v = 0$, that energy is identically zero as evaluated above, and (ORE14), rather than a logarithm, is its receiving equation.

The connection to the original conductor/native chart is explicit. For the actual retained index set I_{sel} , COI11 gives $L_{I_{\text{sel}}} = F_C D_{0I_{\text{sel}}} + L_0 M_{0I_{\text{sel}}}$. NCRP's complete frame is $\mathcal{F} = [F_0, F_C, L_{I_{\text{sel}}}]$, an isomorphism onto the full packet. Put $Y = \mathcal{F}^{-1} I$. Then $\Lambda \mathcal{F} Y = 0$ and the actual kernel metric is exactly

$$H_N = Y^* \mathcal{F}^* G_N \mathcal{F} Y. \quad (\text{ORE19})$$

Expanding $\mathcal{F}^*G_N\mathcal{F}$ retains all nine blocks; expanding its $L_{I_{\text{sel}}}$ block gives all four terms of COI13. Consequently every ratio $d_{K,N}$ in (ORE10)–(ORE18) uses those full blocks. The observation in the analytic native coefficient space is $B_a = Z_a^\top A_{\ell,a}$ of NSO15, with the exact dual metric square stated there. This ordinary transpose is not changed to a star, and this native metric is not substituted for H_N in (ORE19). The relation graph and NSO15 are the connecting maps for the native eight flags; (ORE19) is their complete arithmetic receiving metric.

6 A direct quotient-current receiver with no response division

Put $e_{B,N} = E_N(1 - \theta_N)$ and retain the actual positive denominator and the actual quotient observation numerator

$$D_N^B = \omega_{N+1} + \beta_N(1 - \alpha_N), \quad s_N^B = b_{N+1}^* G_N(I - P_N)v, \quad \Delta_N^B = e_{B,N} - e_{B,N+1} = \frac{|s_N^B|^2}{D_N^B} \geq 0. \quad (\text{ORE20})$$

The last identity is the undivided version of (ORE14), so it remains valid even when $s_N^B = 0$. At the next cutoff (ORE9) says that the first quotient boundary numerator is $\omega_N s_{N-1}^B / D_{N-1}^B$. Substituting this actual numerator into NCE6 yields, for $N \geq q$,

$$\Phi_{B,N} = \frac{2\Re(\overline{s_{N-1}^B} s_N^B)}{D_{N-1}^B}. \quad (\text{ORE21})$$

In particular this computes the exact same original pairing as (ORE11)–(ORE12), without division by either small full-class boundary inner product. The class, observation and metric have not changed. If both drops in (ORE20) are positive, define the actual unit complex phases $z_N = s_N^B / |s_N^B|$ and put $c_N^B = \sqrt{D_N^B / D_{N-1}^B}$. Then

$$\Phi_{B,N} = 2c_N^B \sqrt{\Delta_{N-1}^B \Delta_N^B} \Re(\bar{z}_{N-1} z_N), \quad |\Phi_{B,N}| \leq 2c_N^B \sqrt{\Delta_{N-1}^B \Delta_N^B}. \quad (\text{ORE22})$$

If either drop is zero, (ORE20) makes its numerator zero and (ORE21) gives $\Phi_{B,N} = 0$; no phase is assigned to that zero. Writing $r_{N+1}^B = \det Q_{N+1} / \det Q_N$, (ORE6) further gives

$$(c_N^B)^2 = \frac{\omega_{N+1}}{\omega_N} \frac{r_N^B}{r_{N+1}^B}, \quad \sum_{N=L}^H \Delta_N^B = e_{B,L} - e_{B,H+1}. \quad (\text{ORE23})$$

Thus the coefficient in the undivided current is fixed by the original arithmetic recurrence and original quotient determinants. For any finite interval $q \leq L \leq H$, (ORE22) and $2xy \leq x^2 + y^2$ give the explicit finite bound

$$\sum_{N=L}^H |\Phi_{B,N}| \leq \left(\max_{L \leq N \leq H} c_N^B \right) \left(\Delta_{L-1}^B + 2 \sum_{N=L}^{H-1} \Delta_N^B + \Delta_H^B \right) \leq 2 \left(\max_{L \leq N \leq H} c_N^B \right) e_{B,L-1}.$$

Every coefficient and endpoint is retained. This bound does not assert a new uniform size for c_N^B ; (ORE23) is the exact determinant quantity to calculate. The original individual sign in (ORE22) is precisely the adjacent observed-numerator phase, rather than the phase of the full eigenclass alone.

7 Scope and preceding proofs

The original quotient, boundary and current formulas used here are NCE1–21 in the complete *Native class integration* source; the response energies are JRE1–22. All new equalities above

follow from the full source increment (ORE4) and the displayed attained minimum, with the fixed original I, Λ, v throughout. They reduce the response sequence and its class-energy evolution to the same actual restriction determinants used by the allocation problem. The remaining calculation is the values of those original directional determinants and the residual response phase. Neither the evaluated full action AG29 nor an upper bound for η_N alone assigns the separate projected signs.

Full source derivation and first conductor allocation

8 Original objects, conventions, and the finite minimum

This derivation checks ORE1–23 of `OBSERVATION_RESPONSE_EVOLUTION.tex` directly from the original polynomial source. It also derives further exact receiving identities for equations (5)–(7) of the supplied `LOW_CUTOFF_ENERGIES_CONDUCTOR_AND_MARKED_DEFECT.md`. All adjoints below are conjugate transposes in the stated fixed frames; the inner product is antilinear in its first argument. No original class, source mass, period map, relation, or conductor coordinate is discarded. The calculation is finite and uses the original source domain on which the polynomial norms below are defined.

Keep the actual simple-quartet source, including its original unit, period and logarithm branch, and put

$$\begin{aligned} a &\equiv 1 \pmod{4}, & q &= (a+1)^2, & c &= a/2, & 0 < \delta < 1/2, & \gamma > 2, \\ \chi(S) &= \prod_{u,b=0}^a [S - c - (2u-a)\delta - i(2b-a)\gamma], & E &= \mathbb{C}[S]/(\chi), \\ dm_a(y) &= w_h^{*a}(y) dy, & w_h(y) &= \frac{1}{2\pi} \left| \frac{2\xi}{h} (1/2 + iy) \right|^2, \\ \langle f, g \rangle_{m_a} &= \int_{\mathbb{R}} \overline{f(c+iy)} g(c+iy) dm_a(y). \end{aligned} \tag{A04OID.1}$$

The frame of E is the original increasing-power frame. The original source gives a positive inner product on each finite polynomial space: a nonzero polynomial is nonzero except at finitely many points, and the original source density is nonzero on intervals. The moments used in the original finite source norms are finite. These are the retained source hypotheses, not a replacement Gamma norm.

Write $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and $\mathcal{R}_N = \chi \mathcal{P}_{N-q}$, with $\mathcal{P}_j = 0$ for $j < 0$. For $N \geq q-1$, let $\mathcal{V}_N$ be the orthogonal complement of $\mathcal{R}_N$ in $\mathcal{P}_N$. Polynomial division by the monic χ gives a surjection $\pi_N : \mathcal{P}_N \rightarrow E$ whose kernel is precisely $\mathcal{R}_N$. Consequently $\pi_N|_{\mathcal{V}_N}$ is bijective: its kernel is zero and both spaces have dimension q . Denote its inverse by $T_N : E \rightarrow \mathcal{V}_N$. Every lift of $x \in E$ is uniquely $T_N x + r$ with $r \in \mathcal{R}_N$, and

$$\|T_N x + r\|_{m_a}^2 = \|T_N x\|_{m_a}^2 + \|r\|_{m_a}^2.$$

Thus $T_N x$ is the unique attained minimum over the complete relation space and G_N is exactly its positive Hermitian metric: $x^* G_N y = \langle T_N x, T_N y \rangle_{m_a}$.

Let $p_n(S)$ be the monic source orthogonal polynomial and $\omega_n = \|p_n\|_{m_a}^2 > 0$, and put $b_n = [p_n] \in E$. The polynomials $p_0, \dots, p_N$ form a basis, because their distinct degrees and leading coefficients give a triangular change of basis with diagonal one. Define

$$C_N = [b_0, \dots, b_N], \quad \Omega_N = \text{diag}(\omega_0, \dots, \omega_N), \quad M_N = C_N \Omega_N^{-1} C_N^*.$$

The matrix C_N is surjective, so M_N is positive definite: $z^* M_N z = \sum_{j=0}^N |b_j^* z|^2 / \omega_j$ vanishes only when $C_N^* z = 0$, hence only when $z = 0$. For $x \in E$ the coefficient vector

$$t_N(x) = \Omega_N^{-1} C_N^* M_N^{-1} x$$

satisfies $C_N t_N(x) = x$. If $C_N u = 0$, then $u^* \Omega_N t_N(x) = u^* C_N^* M_N^{-1} x = 0$. Expanding the positive square of $t_N(x) + u$ proves its minimality and yields the explicit, original-source formula

$$G_N = M_N^{-1}, \quad T_N x = \sum_{j=0}^N p_j \frac{b_j^* G_N x}{\omega_j}, \quad G_N^{-1} = \sum_{j=0}^N \frac{b_j b_j^*}{\omega_j}. \tag{A04OID.2}$$

This derives the dual norm and the attained minimum simultaneously.

9 The original source increment and the full boundary action

At one step write, only to display the original factors compactly,

$$G = G_N, \quad G' = G_{N+1}, \quad b = b_{N+1}, \quad \omega = \omega_{N+1}, \quad \beta = b^* G b, \quad D = \omega + \beta.$$

The new summand in (A04OID.2) is exactly bb^*/ω . Direct multiplication, using $b^* G b = \beta$, proves

$$(G')^{-1} = G^{-1} + \frac{bb^*}{\omega}, \quad G' = G - \frac{Gbb^*G}{D}, \quad G'b = rGb, \quad b^*G'b = r\beta, \quad r = \frac{\omega}{D}. \quad (\text{A04OID.3})$$

Indeed the coefficient of Gbb^* in the product of the proposed G' with $G^{-1} + bb^*/\omega$ is $\omega^{-1} - D^{-1} - \beta/(D\omega) = 0$. The resulting inverse identity is the rank-one update recorded by Hager [3, p. 221, equation (2)]; its source column and its positive denominator have been derived here from the actual minimum. For a column x and a row y^* , multilinearity of determinant implies $\det(I + xy^*) = 1 + y^*x$: all terms using two columns from xy^* vanish, and the one-column terms sum to its trace. Applying this identity to (A04OID.3) gives

$$\frac{\det G'}{\det G} = 1 - \frac{\beta}{D} = \frac{\omega}{D} = r. \quad (\text{A04OID.4})$$

We next prove the boundary formula needed for the currents, without importing a conjugation convention from a response formula. Let $A : E \rightarrow E$ be multiplication by S . For $x \in E$, the coefficient of S^N in $T_N x$ is

$$\ell_N x = \frac{b_N^* G_N x}{\omega_N}. \quad (\text{A04OID.5})$$

This follows either from (A04OID.2), since p_N is monic, or by pairing p_N with $T_N x$: the difference $p_N - T_N b_N$ is an old relation and is orthogonal to $T_N x$.

Let Π_N be orthogonal projection onto $\mathcal{V}_N$, acting also on $\mathcal{P}_{N+1}$, and let $J_N = \Pi_N M_y|_{\mathcal{V}_N}$. It is Hermitian, since multiplication by the real variable y is Hermitian in the original integral. Set $\widehat{b}_j = T_N b_j$. For $f = T_N x$, $Sf - (\ell_N x)p_{N+1}$ has degree at most N and class $Ax - (\ell_N x)b_{N+1}$. Since $p_{N+1} \perp \mathcal{P}_N$, projecting this polynomial onto $\mathcal{V}_N$ gives

$$T_N A T_N^{-1} = cI + iJ_N + \frac{\widehat{b}_{N+1} \widehat{b}_N^*}{\omega_N}, \quad (\text{A04OID.6})$$

$$D_N := A^* G_N + G_N A - aG_N = \frac{G_N b_{N+1} b_N^* G_N + G_N b_N b_{N+1}^* G_N}{\omega_N}.$$

Here the star in the source-space rank-one operator denotes its actual source inner product; the matrix identity on E uses the original power frame. The sign is positive because the class $(\ell_N x)b_{N+1}$ is added back after the degree reduction.

Retain the full original eigenclass

$$\lambda = c + d + it, \quad d = a\delta > 0, \quad t = a\gamma > 0, \quad v = v_\lambda = \frac{\chi(S)}{(S - \lambda)\chi'(\lambda)}, \quad Av = \lambda v, \quad (\text{A04OID.7})$$

including all its original low, conductor and native coordinates. Put $f = T_N v$, $E_N = v^* G_N v$, and $\mu_N = E_N^{-1} \int y |f(c + iy)|^2 dm_a(y) \in \mathbb{R}$. The polynomial $(S - \lambda)f$ is a relation. If $r_N(S) = \chi(S)S^{N+1-q}$, its difference from $(\ell_N v)r_N$ is an old relation. Moreover $p_{N+1} - r_N \in \mathcal{P}_N$ represents b_{N+1} , so

$$\langle f, r_N \rangle = -\langle f, p_{N+1} - r_N \rangle = -v^* G_N b_{N+1}.$$

Consequently

$$E_N[-d + i(\mu_N - t)] = \langle f, (S - \lambda)f \rangle = -\frac{\overline{b_{N+1}^*} G_N v b_N^* G_N v}{\omega_N}.$$

Taking the conjugate, with the displayed minus sign retained, proves

$$\mathbf{c}_N := \frac{\overline{b_N^* G_N v} b_{N+1}^* G_N v}{\omega_N E_N} = d + i(\mu_N - t), \quad \Re \mathbf{c}_N = d > 0. \quad (\text{A04OID.8})$$

Both boundary inner products are therefore nonzero. In particular $b \neq 0$, $\beta > 0$, and $0 < r < 1$ in (A04OID.3).

The original measure is even and $\chi(a - S) = (-1)^q \chi(S)$, directly from the paired full root grid. The isometry $f(S) \mapsto f(a - S)$ preserves both the relation space and its orthogonal complement. Uniqueness of the monic orthogonal polynomial gives $p_j(a - S) = (-1)^j p_j(S)$, so the attained columns $\widehat{b}_N, \widehat{b}_{N+1}$ have opposite parities. Hence

$$b_N^* G_N b_{N+1} = 0. \quad (\text{A04OID.9})$$

The squared norm of the projection of v onto these two nonzero orthogonal columns is

$$\frac{|b_N^* G_N v|^2}{b_N^* G_N b_N} + \frac{|b_{N+1}^* G_N v|^2}{b_{N+1}^* G_N b_{N+1}} \leq E_N.$$

Thus the original two boundary energies satisfy

$$p_{1,N} := \frac{|b_N^* G_N v|^2}{E_N b_N^* G_N b_N} > 0, \quad p_{2,N} := \frac{|b_{N+1}^* G_N v|^2}{E_N b_{N+1}^* G_N b_{N+1}} > 0, \quad p_{1,N} + p_{2,N} \leq 1. \quad (\text{A04OID.10})$$

In particular each energy is strictly below one on this original eigenclass domain.

10 The original observation minimum and its exact update

Fix the full original observation $\Lambda : E \rightarrow B = \text{im } \Lambda$ in a fixed frame of B , retaining its period map. Let $K = \ker \Lambda$ and let $I : \mathbb{C}^{\dim K} \rightarrow E$ be any fixed injective frame of K . At the present cutoff define

$$H = I^* G I, \quad P = I H^{-1} I^* G, \quad Q = (\Lambda G^{-1} \Lambda^*)^{-1}, \quad L = G^{-1} \Lambda^* Q : B \rightarrow E. \quad (\text{A04OID.11})$$

Surjectivity of Λ makes $\Lambda G^{-1} \Lambda^*$ positive definite. The identity $P^2 = P$, its image K , and $P^* G = G P$ follow by multiplication. Also $\Lambda L = I_B$ and $I^* G L = I^* \Lambda^* Q = 0$. Every lift of $y \in B$ is uniquely $Ly + k$ with $k \in K$ and its squared norm is $y^* Q y + k^* G k$. Thus L is the actual unique minimum lift and

$$L \Lambda = I_E - P, \quad G(I_E - P) = \Lambda^* Q \Lambda. \quad (\text{A04OID.12})$$

For $K = 0$ the corresponding inverse in H is empty and $P = 0$; for $B = 0$ the corresponding inverse in Q is empty. Empty determinants equal one throughout.

Keep

$$\alpha = \frac{b^* G P b}{\beta} \in [0, 1], \quad \xi = 1 - r = \frac{\beta}{D}, \quad d_K = \frac{\det H'}{\det H}, \quad u = \Lambda b.$$

Restriction of (A04OID.3) to I and the proved determinant identity give

$$H' = H - \frac{z z^*}{D}, \quad z = I^* G b, \quad z^* H^{-1} z = \beta \alpha,$$

and hence

$$d_K = 1 - \xi \alpha, \quad r \leq d_K \leq 1, \quad D^B := \omega + \beta(1 - \alpha) = D d_K > 0. \quad (\text{A04OID.13})$$

The quotient inverse updates by the actual observed column:

$$(Q')^{-1} = Q^{-1} + \frac{u u^*}{\omega}, \quad u^* Q u = b^* G (I - P) b = \beta(1 - \alpha).$$

Multiplication as in (A04OID.3) now proves

$$Q' = Q - \frac{Quu^*Q}{D^B}, \quad r^B := \frac{\det Q'}{\det Q} = \frac{\omega}{D^B} = \frac{r}{d_K}. \quad (\text{A04OID.14})$$

In particular the exact nonnegative logarithmic allocations are $-\log d_K$ and $-\log r + \log d_K$, with sum $-\log r$.

There is also an exact update of the original minimum section and of the projector itself. Multiplying the two matrices in $L' = (G^{-1} + bb^*/\omega)\Lambda^*Q'$ and using $\omega + u^*Qu = D^B$ gives

$$L' = L + \frac{Pbu^*Q}{D^B}, \quad P' = P - \frac{Pbb^*G(I - P)}{D^B}. \quad (\text{A04OID.15})$$

The second identity follows by multiplying the first by Λ and applying (A04OID.12). Its correction is an actual map into the original kernel; it does not replace that kernel by a rank. It is defined even when $u = 0$ or $\Lambda v = 0$.

11 Response transport, conjugations, and energy transport

Return cutoff indices to every quantity and retain the original rows

$$U_N = \frac{b_N^* G_N P_N v}{b_N^* G_N v}, \quad V_N = \frac{b_{N+1}^* G_N P_N v}{b_{N+1}^* G_N v}, \quad \theta_N = \frac{v^* G_N P_N v}{E_N}. \quad (\text{A04OID.16})$$

They have their original order. Write $s_N = b_{N+1}^* G_N v$ and $s_N^B = b_{N+1}^* G_N (I - P_N) v = (1 - V_N) s_N$. From (A04OID.3), the next full numerator for this same b_{N+1} is $r_{N+1} s_N$. From (A04OID.14), its next quotient numerator is

$$b_{N+1}^* G_{N+1} (I - P_{N+1}) v = u^* Q_{N+1} \Lambda v = \frac{\omega_{N+1}}{D_N^B} s_N^B = \frac{r_{N+1}}{d_{K,N}} s_N^B. \quad (\text{A04OID.17})$$

Dividing by the proved nonzero full numerator yields

$$\boxed{U_{N+1} = 1 - \frac{1 - V_N}{d_{K,N}}}. \quad (\text{A04OID.18})$$

No conjugation occurs in this linear transport. Conjugation enters only when the earlier row is used in a Hermitian current.

For $x_{K,N} = P_N v$ and $x_{B,N} = (I - P_N) v$, define the actual centered pairings

$$\Phi_{K,N} = x_{K,N}^* D_N x_{K,N}, \quad \Phi_{B,N} = x_{B,N}^* D_N x_{B,N}, \quad \Phi_{\times,N} = 2\Re(x_{K,N}^* D_N x_{B,N}).$$

Equation (A04OID.6) gives, with every orientation explicit,

$$\begin{aligned} P_{K,N}^{\text{resp}} &= \overline{U}_N V_N, & P_{B,N}^{\text{resp}} &= (1 - \overline{U}_N)(1 - V_N), \\ P_{\times,N}^{\text{resp}} &= \overline{U}_N + V_N - 2\overline{U}_N V_N, & \sum_X P_{X,N}^{\text{resp}} &= 1, \\ \Phi_{X,N} &= 2E_N \Re(\mathbf{c}_N P_{X,N}^{\text{resp}}), & \sum_X \Phi_{X,N} &= 2dE_N. \end{aligned} \quad (\text{A04OID.19})$$

For example its quotient term is $2\Re(\overline{b_N^* G_N x_{B,N}} b_{N+1}^* G_N x_{B,N})/\omega_N$. Its two cross terms are $2\Re(\overline{b_N^* G_N x_{K,N}} b_{N+1}^* G_N x_{B,N} + \overline{b_N^* G_N x_{B,N}} b_{N+1}^* G_N x_{K,N})/\omega_N$. These expansions prove the three products, including their conjugations. For $N \geq q$, put $R_N = 1 - V_N$. Then (A04OID.18) gives

$$P_{B,N}^{\text{resp}} = \frac{\overline{R_{N-1}} R_N}{d_{K,N-1}}, \quad P_{K,N}^{\text{resp}} = \left(1 - \frac{\overline{R_{N-1}}}{d_{K,N-1}}\right) (1 - R_N). \quad (\text{A04OID.20})$$

At $N = q - 1$ the original U_{q-1} is retained; there is no earlier surjective quotient source of dimension q to insert.

Set $\eta_N = (1 - r_{N+1})p_{2,N}$. Substitution into (A04OID.3), followed by (A04OID.10), proves

$$\begin{aligned} E_{N+1} &= E_N - \frac{|s_N|^2}{\omega_{N+1} + \beta_N}, & \frac{E_{N+1}}{E_N} &= 1 - \eta_N, \\ 0 < \eta_N < 1 - r_{N+1} < 1, & r_{N+1} < \frac{E_{N+1}}{E_N} < 1. \end{aligned} \quad (\text{A04OID.21})$$

The strict comparison with r_{N+1} strengthens the non-strict Cauchy–Schwarz bound in ORE13; it follows from the other nonzero, orthogonal boundary column, not from an assumed asymptotic profile. For the observed energy $e_{B,N} = E_N(1 - \theta_N) = (\Lambda v)^* Q_N \Lambda v$, (A04OID.14) yields

$$\Delta_N^B := e_{B,N} - e_{B,N+1} = \frac{|s_N^B|^2}{D_N^B} \geq 0, \quad (1 - \eta_N)(1 - \theta_{N+1}) = 1 - \theta_N - \frac{\eta_N |1 - V_N|^2}{d_{K,N}}. \quad (\text{A04OID.22})$$

Rearranging the second equation, without dividing by $1 - \theta_N$, gives

$$\theta_{N+1} - \theta_N = \frac{\eta_N}{1 - \eta_N} \left[\theta_N - 1 + \frac{|1 - V_N|^2}{d_{K,N}} \right], \quad 0 \leq \frac{\eta_N |1 - V_N|^2}{d_{K,N}} \leq 1 - \theta_N. \quad (\text{A04OID.23})$$

If $\Lambda v \neq 0$, positive definiteness of Q_{N+1} makes the last upper inequality strict. If $\Lambda v = 0$, $v \in K$ and $P_N v = v$, so $\theta_N = 1$, $U_N = V_N = 1$, and every observed energy and observed drop is zero. This evaluates that stratum without an undefined logarithm or phase.

Finally (A04OID.3) gives, for the same column b_{N+1} ,

$$p_{1,N+1} = \frac{r_{N+1} p_{2,N}}{1 - (1 - r_{N+1}) p_{2,N}}. \quad (\text{A04OID.24})$$

Both numerator and denominator use the retained class and original source metric at their displayed cutoffs.

12 The quotient-current receiver and its finite bound

For $N \geq q$, (A04OID.17) says that the actual first quotient numerator at cutoff N is

$$b_N^* G_N (I - P_N) v = \frac{\omega_N}{D_{N-1}^B} s_{N-1}^B.$$

Insert this identity, with its real positive coefficient, into the quotient expansion following (A04OID.19). The factor ω_N cancels its matching boundary denominator, proving

$$\boxed{\Phi_{B,N} = \frac{2\Re(\overline{s_{N-1}^B} s_N^B)}{D_{N-1}^B}}. \quad (\text{A04OID.25})$$

This is precisely ORE21. The conjugate belongs to s_{N-1}^B ; neither replacing $\overline{s_{N-1}^B} s_N^B$ by $s_{N-1}^B s_N^B$ nor replacing D_{N-1}^B by D_N^B preserves the displayed derivation. It agrees exactly with (A04OID.20), because $s_N^B = (1 - V_N) s_N$ and $b_N^* G_N v = r_N s_{N-1}$.

When both drops are positive, set $z_N = s_N^B / |s_N^B|$ and $c_N^B = \sqrt{D_N^B / D_{N-1}^B} > 0$. Equation (A04OID.22) then proves

$$\Phi_{B,N} = 2c_N^B \sqrt{\Delta_{N-1}^B \Delta_N^B} \Re(\bar{z}_{N-1} z_N), \quad |\Phi_{B,N}| \leq 2c_N^B \sqrt{\Delta_{N-1}^B \Delta_N^B}. \quad (\text{A04OID.26})$$

If either drop is zero its numerator is zero and (A04OID.25) gives $\Phi_{B,N} = 0$; a phase at zero is unnecessary. Since $D_N^B = \omega_{N+1}/r_{N+1}^B$,

$$(c_N^B)^2 = \frac{\omega_{N+1}}{\omega_N}, \frac{r_N^B}{r_{N+1}^B}, \quad \sum_{N=L}^H \Delta_N^B = e_{B,L} - e_{B,H+1}. \quad (\text{A04OID.27})$$

For a finite interval $q \leq L \leq H$, applying $2\sqrt{xy} \leq x + y$ to each term of (A04OID.26) gives

$$\begin{aligned} \sum_{N=L}^H |\Phi_{B,N}| &\leq \max_{L \leq N \leq H} c_N^B \left(\Delta_{L-1}^B + 2 \sum_{N=L}^{H-1} \Delta_N^B + \Delta_H^B \right) \\ &\leq 2 \max_{L \leq N \leq H} c_N^B, e_{B,L-1}. \end{aligned} \quad (\text{A04OID.28})$$

The second inequality follows because all drops are nonnegative and their sum from $L-1$ to H is $e_{B,L-1} - e_{B,H+1} \leq e_{B,L-1}$. This proves ORE22–23 and their endpoint coefficients exactly. It does not supply a uniform upper bound for the displayed c_N^B .

13 A further exact relation: the next kernel overlap determines the present quotient determinant

Define the next first-boundary kernel overlap in its original metric,

$$\pi_{1,K,N+1} = \frac{b_{N+1}^* G_{N+1} P_{N+1} b_{N+1}}{b_{N+1}^* G_{N+1} b_{N+1}} \in [0, 1]. \quad (\text{A04OID.29})$$

In the one-step notation, $z = I^* G b$ and $H' = H - z z^* / D$. Multiplication proves

$$(H')^{-1} = H^{-1} + \frac{H^{-1} z z^* H^{-1}}{D^B}, \quad z^* (H')^{-1} z = \beta \alpha + \frac{(\beta \alpha)^2}{D^B} = \frac{\beta \alpha}{d_K}.$$

As $I^* G' b = r I^* G b$, the numerator in (A04OID.29) is $r^2 \beta \alpha / d_K$, while its denominator is $r \beta$. Therefore, with $\pi = \pi_{1,K,N+1}$,

$$\boxed{\pi = \frac{r \alpha}{d_K}, \quad \alpha = \frac{\pi}{r + (1-r)\pi}, \quad d_K = \frac{r}{r + (1-r)\pi}, \quad r^B = r + (1-r)\pi.} \quad (\text{A04OID.30})$$

All denominators are positive since $r > 0$. For $K = 0$ these give $\pi = \alpha = 0, d_K = 1, r^B = r$; for $K = E$ they give $\pi = \alpha = 1, d_K = r, r^B = 1$. Thus the endpoint conventions are included in the same exact identity.

This relation is additional to ORE1–23. It makes a bound on the actual next first-boundary kernel overlap into a bound on the same step's quotient determinant. In particular the source's conductor bound for $\pi_{1,K,q}$ has this exact receiving map at the first source increment. It does not require a choice of kernel basis or a replacement by the dimension of that kernel.

14 The complete coefficient chart and both original windows

For completeness the metric update above retains the full arithmetic coordinate map. Write $\chi(S) = \sum_{\ell=0}^q c_\ell S^\ell$, $c_q = 1$, and define the residue-dual polynomials

$$\mathfrak{p}_n(S) = \sum_{\ell=n+1}^q c_\ell S^{\ell-n-1}, \quad 0 \leq n < q, \quad F e_n = \mathfrak{p}_n.$$

Their distinct monic degrees $q-1-n$ prove that $F : \mathbb{C}^q \rightarrow E$ is invertible. With $\mathbb{V}_{n,\alpha} = \beta_\alpha^n$ and $\mathcal{L}(\nu) = \sum_\alpha \nu_\alpha \chi(S) / (S - \beta_\alpha)$, the residue pairing gives $F = \mathcal{L} \mathbb{V}^{-1}$. To see the pairing directly, the

sum of finite residues of $\mathfrak{p}_n(S)S^j/\chi(S)$ is its S^{-1} coefficient at infinity. It is zero for $j < n$, one for $j = n$, and zero for $j > n$, the latter from $S^{n+1}\mathfrak{p}_n = \chi - \sum_{\ell=0}^n c_\ell S^\ell$. This proves the claimed inverse moment map with its ordinary Vandermonde transpose convention.

Use $v_A = \text{ord}_0 E_A$ for the conductor vanishing order, to avoid confusing it with the retained class v . Put $h_A = q - v_A$ and $d_A = (a - 7)^2$, on the original conductor domain. Let P_{v_A} insert the initial v_A zero coordinates. The actual conductor map $C_a^\top$ satisfies

$$\mathbb{V}C_a^\top = P_{v_A}J, \quad J_{i,\alpha'} = \sum_{r=0}^i \binom{v_A+i}{r} \mu_{v_A+i-r}(\beta'_{\alpha'})^r.$$

This is the ordinary product rule for $E_A(z) \sum_{\alpha'} \eta_{\alpha'} e^{\beta'_{\alpha'} z}$. The map $C_a^\top$ is injective: if this entire product is zero, the nonzero analytic factor E_A is nonzero on an open disc, so the exponential polynomial vanishes identically; its first d_A derivatives give an invertible Vandermonde and force $\eta = 0$. Thus J has rank d_A . Retain an actual selected minor I_{sel} with $J_{I_{\text{sel}}}$ invertible and complement $J(I_{\text{sel}})$. Put

$$H_{I_{\text{sel}}} = J_{I_{\text{sel}}}^{-1} E_{I_{\text{sel}}}^\top, \quad Q_{I_{\text{sel}}} = E_{J(I_{\text{sel}})}^\top - J_{J(I_{\text{sel}})} H_{I_{\text{sel}}}.$$

Checking the selected and complementary rows separately proves $JH_{I_{\text{sel}}} + E_{J(I_{\text{sel}})} Q_{I_{\text{sel}}} = I$. It follows that the full chart

$$\mathcal{F} = [F_0, F_C, L_{I_{\text{sel}}}], \quad F_0 = F E_{\{0, \dots, v_A-1\}}, \quad F_C = F P_{v_A} J = \mathcal{L} C_a^\top, \quad L_{I_{\text{sel}}} = F P_{v_A} E_{J(I_{\text{sel}})}$$

is invertible. Its inverse retains all three coordinates, including $H_{I_{\text{sel}}} P_{v_A}^\top F^{-1} x$ in the conductor block. For the consecutive chart 0, application of its H_0, Q_0 to the selected coefficient columns proves the complete transition

$$L_{I_{\text{sel}}} = F_C D_{0I_{\text{sel}}} + L_0 M_{0I_{\text{sel}}}, \quad D_{0I_{\text{sel}}} = H_0 E_{J(I_{\text{sel}})}, \quad M_{0I_{\text{sel}}} = Q_0 E_{J(I_{\text{sel}})}.$$

No change of the observation frame I is implicit in this native section notation. Set $Y = \mathcal{F}^{-1}I$. Then

$$\Lambda \mathcal{F} Y = 0, \quad H_N = Y^* \mathcal{F}^* G_N \mathcal{F} Y. \quad (\text{A04OID.31})$$

If $Y = (Y_0, Y_C, Y_I)^\top$, the latter is explicitly the sum of all nine terms $\sum_{r,s \in \{0,C,I\}} Y_r^* F_r^* G_N F_s Y_s$. For example the native diagonal block under its transition is

$$D^* F_C^* G_N F_C D + D^* F_C^* G_N L_0 M + M^* L_0^* G_N F_C D + M^* L_0^* G_N L_0 M.$$

Thus ORE19 retains every cross metric and every conductor term. Any analytic-native formula using an ordinary transpose remains in its own coefficient maps; this Hermitian pullback does not change that transpose to an adjoint.

For every positive scalar sequence f_N , cancellation of consecutive ratios proves

$$\log \frac{f_{2q-1} f_{2q}}{f_{q-1} f_q} = \sum_{N=q-1}^{2q-1} w_N \log \frac{f_{N+1}}{f_N}, \quad w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \quad (q \leq N \leq 2q-2). \quad (\text{A04OID.32})$$

Indeed the interior exponent of each f_j is $w_{j-1} - w_j$; the only surviving exponents are $-1, -1, +1, +1$ at $q-1, q, 2q-1, 2q$. Use this with $\det G_N, \det H_N, \det Q_N, E_N$ and the increments $r_{N+1}, d_{K,N}, r_{N+1}/d_{K,N}, 1 - \eta_N$. It also applies to $e_{B,N}$ when $\Lambda v \neq 0$. When $\Lambda v = 0$, that sequence is identically zero and its receiving equation is (A04OID.22), not a logarithm. This proves ORE18 with its original two windows and both endpoints.

15 Exact low-cutoff evaluation, with all complex phases

The supplied low-cutoff note uses k for the same tensor degree called a above. In this section set $k = a$ and preserve the note's $q = (k + 1)^2$, $c = k/2$, $d = k\delta$, $t = k\gamma$ and

$$z = \frac{\lambda - c}{i} = t - id, \quad Q(y) = i^{-q}\chi(c + iy), \quad b_\lambda = \chi'(\lambda)^{-1}.$$

The full physical polynomial Q is real and has parity $(-1)^q$. Retain the original source integrals and their original masses:

$$\begin{aligned} \nu_0 &= \int Q(y)^2 dm_k(y), & \nu_1 &= \int y^2 Q(y)^2 dm_k(y), & d\mu_Q(y) &= \nu_0^{-1} Q(y)^2 dm_k(y), \\ a_z &= \int \frac{d\mu_Q(y)}{y - z}, & m_z &= \int \frac{d\mu_Q(y)}{|y - z|^2}, & s_2 &= \frac{\nu_1}{\nu_0}, & d_0 &= \frac{\omega_q}{\nu_0}, & d_1 &= \frac{\omega_{q+1}}{\nu_1}, \\ M &= m_z - |a_z|^2, & g &= 1 + za_z. \end{aligned} \tag{A04OID.33}$$

The probability μ_Q is only the note's auxiliary device for ratios: every norm below retains the multiplying ν_0 or ν_1 . The symbols d_0, d_1 here are the two monic norm ratios of its equation (4); they are not the unrelated tail coefficient $d_0(h)$ mentioned later in that note.

At $N = q - 1$ no relation occurs in the minimum fibre, and its actual polynomial representative is

$$f_{q-1}(c + iy) = b_\lambda i^{q-1} \frac{Q(y)}{y - z}, \quad E_{q-1} = |b_\lambda|^2 \nu_0 m_z.$$

Its leading coefficient in S is b_λ , so $F_{13,q-1} = b_\lambda \omega_{q-1}$ and $F_{11,q-1} = \omega_{q-1}$. The representative of b_q is $p_q - \chi$. Orthogonality gives

$$\|p_q - \chi\|^2 = \nu_0 - \omega_q, \quad \langle p_q - \chi, f_{q-1} \rangle = -\langle \chi, f_{q-1} \rangle.$$

Since $\overline{iq}i^{q-1} = i^{-1} = -i$, the latter numerator is exactly $ib_\lambda \nu_0 a_z$, including its phase. Thus

$$\begin{aligned} F_{13,q-1} &= b_\lambda \omega_{q-1}, & F_{23,q-1} &= ib_\lambda \nu_0 a_z, & F_{22,q-1} &= \nu_0 - \omega_q, \\ p_{1,q-1} &= \frac{\omega_{q-1}}{\nu_0 m_z}, & p_{2,q-1} &= \frac{|a_z|^2}{(1 - d_0)m_z}, & r_q &= d_0. \end{aligned} \tag{A04OID.34}$$

The first two energy formulas rederive the note's equation (5).

At $N = q$ the relation space is $\mathbb{C}\chi$. Its projection coefficient is $\langle \chi, f_{q-1} \rangle / \nu_0 = -ib_\lambda a_z$. Subtracting this entire original relation gives

$$f_q(c + iy) = b_\lambda i^{q-1} Q(y) \left(\frac{1}{y - z} - a_z \right), \quad E_q = |b_\lambda|^2 \nu_0 M. \tag{A04OID.35}$$

This is the note's equation (6), obtained by the actual orthogonal projection. Its first boundary numerator is, by the already proved update, $F_{13,q} = d_0 F_{23,q-1} = ib_\lambda \omega_q a_z$.

The next monic relation is $R(S) = \chi(S)(S - c)$. It is orthogonal to χ , because μ_Q is even, and its norm is ν_1 . Both p_{q+1} and R have parity $(-1)^{q+1}$ about c ; therefore $p_{q+1} - R$ is orthogonal to χ and is the attained representative of b_{q+1} at cutoff q . Its norm is $\nu_1 - \omega_{q+1}$. Moreover

$$\int \frac{y}{y - z} d\mu_Q(y) = 1 + za_z = g, \quad \overline{i^{q+1}}i^{q-1} = i^{-2} = -1.$$

It follows that

$$\begin{aligned} F_{13,q} &= ib_\lambda \omega_q a_z, & F_{23,q} &= b_\lambda \nu_0 g, & F_{22,q} &= \nu_1 - \omega_{q+1}, & r_{q+1} &= d_1, \\ p_{1,q} &= \frac{d_0 |a_z|^2}{(1 - d_0)M}, & p_{2,q} &= \frac{|g|^2}{s_2(1 - d_1)M}. \end{aligned} \tag{A04OID.36}$$

This proves the note's equation (7) and its original phases. The nonzero boundary numerators proved in (A04OID.8) give $0 < d_0, d_1 < 1$.

The same exact source update now determines further finite quantities:

$$\boxed{\eta_{q-1} = \frac{|a_z|^2}{m_z}, \quad \eta_q = \frac{|g|^2}{s_2 M}, \quad E_{q+1} = |b_\lambda|^2 \nu_0 \left(M - \frac{|g|^2}{s_2} \right).} \quad (\text{A04OID.37})$$

For direct verification, at cutoff $q+1$ the entire relation space is the orthogonal span of χ and R , and

$$f_{q+1}(c + iy) = b_\lambda i^{q-1} Q(y) \left(\frac{1}{y-z} - a_z - \frac{g}{s_2} y \right).$$

Its original norm is the last expression in (A04OID.37). It is strictly positive because $1/(y-z)$ cannot equal an affine function on a set containing an interval: multiplying such an identity by $y-z$ would give a polynomial identity whose quadratic and linear coefficients force both affine coefficients to be zero, contradicting its constant coefficient one. This also proves the strict inequalities $M > 0$ and $M - |g|^2/s_2 > 0$ by their respective orthogonal residuals. There is no missing new minimum in (A04OID.37).

The original full currents consequently have the explicit phases

$$\mathbf{c}_{q-1} = \frac{ia_z}{m_z}, \quad \mathbf{c}_q = \frac{-i\overline{a_z}g}{M}, \quad \boxed{\mathbf{c}_q = \frac{\overline{\mathbf{c}_{q-1}} - \eta_{q-1}(d + it)}{1 - \eta_{q-1}}.} \quad (\text{A04OID.38})$$

Indeed $\Im a_z = -dm_z$ by the integrand $(y-z)^{-1} = (y-t-id)/|y-z|^2$, so the first current has real part d . Also $\Im(\overline{a_z}g) = -\Im a_z + |a_z|^2 \Im z = dM$, which proves the second current has real part d . For its transport formula use $a_z = -im_z \mathbf{c}_{q-1}$, $|a_z|^2/m_z = \eta_{q-1}$, and $iz = d + it$. Equivalently, the original source means satisfy the exact relation

$$\mu_q = 2t - \frac{\mu_{q-1}}{1 - \eta_{q-1}}. \quad (\text{A04OID.39})$$

It follows by comparing imaginary parts in (A04OID.38) and (A04OID.8). These equations contain the original physical center and imaginary coordinate; no phase is normalized away.

For the original observation, write $\kappa_0 = d_{K,q-1}$ and $\kappa_1 = d_{K,q}$ only in the following low-cutoff formulas. As $D_{q-1} = \nu_0$ and $D_q = \nu_1$,

$$\begin{aligned} D_{q-1}^B &= \nu_0 \kappa_0, & D_q^B &= \nu_1 \kappa_1, & (c_q^B)^2 &= s_2 \frac{\kappa_1}{\kappa_0}, \\ \Delta_{q-1}^B &= \frac{|b_\lambda|^2 \nu_0 |a_z|^2 |1 - V_{q-1}|^2}{\kappa_0}, & \Delta_q^B &= \frac{|b_\lambda|^2 \nu_0 |g|^2 |1 - V_q|^2}{s_2 \kappa_1}, \\ \boxed{\Phi_{B,q} &= \frac{2|b_\lambda|^2 \nu_0}{\kappa_0} \Re \left[-i\overline{a_z}g(1 - \overline{V}_{q-1})(1 - V_q) \right].} \end{aligned} \quad (\text{A04OID.40})$$

Every equation follows by inserting the explicitly computed complex numerators into (A04OID.22) and (A04OID.25). It is a concrete receiving map for the note's equations (5)–(7); it preserves the still actual period-dependent responses and restriction determinants. It does not assign their signs from a full-class energy. The additional determinant relation (A04OID.30) is especially explicit at this first step:

$$\boxed{r_q^B = d_0 + (1 - d_0)\pi_{1,K,q}, \quad \kappa_0 = \frac{d_0}{d_0 + (1 - d_0)\pi_{1,K,q}}.} \quad (\text{A04OID.41})$$

16 The actual conductor evaluates the first logarithmic allocation

The domain of this section is the original simple-packet conductor domain, $k \equiv 1 \pmod{4}$, $k \geq 9$, and its fixed actual period ϖ , with the original finite root-comparison guards retained where the LRS estimates below are used. Its full observation is the one in inherited FEX1–3 [6]; in particular its proved conductor identity is

$$K = \ker \Lambda_k \subset \ker C, \quad C[P] = ((\mathcal{T}_A P)(\beta'_{\alpha'}))_{\alpha'}, \quad \chi_{k-8} \mid \mathcal{T}_A P \quad ([P] \in K), \quad (\text{A04OID.42})$$

where all lower roots $\beta'_{\alpha'}$ are retained. The exact receiving map from the original observation is $\overline{C} : B \rightarrow \mathbb{C}^{q'}$, defined by $\overline{C}(\Lambda_k x) = Cx$. It is well-defined because $\ker \Lambda_k \subset \ker C$, is linear, and is the unique map with $C = \overline{C}\Lambda_k$ because Λ_k is onto B . Write

$$q' = (k-7)^2, \quad c' = c-4, \quad \beta_{8,r,s} = 4 + (2r-8)\delta + i(2s-8)\gamma, \quad 0 \leq r, s \leq 8,$$

and retain the complete conductor

$$\begin{aligned} (\mathcal{T}_A P)(S') &= \sum_{r,s=0}^8 a_{rs} P(S' + \beta_{8,r,s}), \quad \mu_j = \sum_{r,s=0}^8 a_{rs} \beta_{8,r,s}^j, \\ v_A &= \min\{j : \mu_j \neq 0\} \leq 80, \quad \mathcal{C}_A = \frac{v_A!}{\mu_{v_A}} \mathcal{T}_A. \end{aligned} \quad (\text{A04OID.43})$$

This is the original displayed conductor scalar. It does not change the arithmetic source or the observation. Taylor expansion of each shift is finite on a polynomial, so $\mathcal{C}_A = \sum_{j \geq v_A} v_A! \mu_j \partial_S^j / (\mu_{v_A} j!)$. Therefore on a polynomial of degree $N \geq v_A$ its leading coefficient is exactly $(N)_{v_A}$ times the original leading coefficient.

The divisibility in (A04OID.42) holds for every representative of the same class, including the attained one. Indeed at a lower root the shifted point $\beta'_{u',w'} + \beta_{8,r,s} = \beta_{k,u'+r,w'+s}$ is an original upper root. Thus every summand of $\mathcal{T}_A(\chi H)(\beta'_{u',w'})$ is zero, for every polynomial H . Simplicity of all lower roots then proves $\chi_{k-8} \mid \mathcal{T}_A(\chi H)$. This verifies the exact map on quotient classes rather than assuming its invariance under the minimum relation.

Let σ be the original order-one Gamma comparison measure, in the physical coordinate $S = c + iy$ (and $S' = c' + iy$ in the target). Its complete density and monic norms are

$$d\sigma(y) = \frac{1}{2\pi} |\Gamma(1/4 + iy/2)|^2 dy, \quad \gamma_n = \sqrt{2\pi} n! (1/2)_n = \sqrt{2\pi} \frac{\Gamma(2n+1)}{4^n}.$$

Retain the original whole-polynomial comparison, valid through degree $2q$,

$$\ell_k \|P\|_{\sigma,c}^2 \leq \|P\|_{m_k}^2 \leq u_k \|P\|_{\sigma,c}^2, \quad \log(u_k/\ell_k) = O_h(k + \log q). \quad (\text{A04OID.44})$$

This comparison is applied before minimization. It is not a pointwise lower bound for the arithmetic density.

For clarity, the forward conductor bound of FEX9–11 is reproduced in the order-one source needed here. In its original monic Gamma basis g_n ,

$$\sum_{n \geq 0} \frac{g_n(y)}{n!} z^n = (1+z^2)^{-1/4} e^{y \arctan z}, \quad w_n = \frac{n!}{(1/2)_n}, \quad \varphi_n = g_n / \sqrt{\gamma_n}.$$

The displayed generating identity is also exactly DLMF 18.23.7 [5] with $g_n(y) = n! P_n^{(1/4)}(y/2; \pi/2)$: its two factors $(1-iz)^{-1/4+iy/2} (1+iz)^{-1/4-iy/2}$ multiply to the displayed right side. This identifies the physical coordinate and monic coefficient without altering the source norm. Differentiating the coefficient identity and using $\arctan z = \sum_{j \geq 1, j \text{ odd}} (-1)^{(j-1)/2} z^j / j$ gives

$$\partial_y \varphi_n = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \frac{(-1)^{(j-1)/2}}{j} \sqrt{\frac{w_n}{w_{n-j}}} \varphi_{n-j}.$$

The source mass cancels from these coefficient ratios only. For $n > m \geq 1$, the product of $j(j-1) \leq (j-1/2)^2$ gives $w_n/w_m \leq \sqrt{n/m}$; the same product with $w_1 = 2$ gives $w_n \leq 2\sqrt{n}$. In the absolute derivative matrix use weights $w_n^{-1/2}$. Its weighted row sums are at most $H_N = \sum_{j=1}^N 1/j$. Its weighted column sum at n is at most $H_n + 4$, since for $1 \leq m < n$

$$\frac{\sqrt{n/m} - 1}{n - m} = \frac{1}{\sqrt{m}(\sqrt{n} + \sqrt{m})} \leq \frac{1}{\sqrt{nm}}, \quad \sum_{m=1}^{n-1} \frac{1}{\sqrt{nm}} \leq 2,$$

and the $m = 0$ term is at most $2/\sqrt{n} \leq 2$. Weighted Cauchy–Schwarz on each row, then summation of the column bound, proves $\|\partial_y\| \leq \sqrt{H_N(H_N + 4)} \leq H_N + 2$. The finite exponential of this nilpotent derivative is translation, so $\|f(y+z)\|_\sigma \leq e^{|z|(H_N+2)} \|f\|_\sigma$. Since the original center displacement is four, triangle inequality in (A04OID.43) proves the full retained bound

$$\begin{aligned} \|\mathcal{C}_A P\|_{\sigma, c-4} &\leq M_N \|P\|_{\sigma, c}, \\ M_N &= \max \left\{ 1, \frac{v_A!}{|\mu_{v_A}|} \sum_{r,s=0}^8 |a_{rs}| e^{|\beta_{8,r,s-4}|(H_N+2)} \right\}. \end{aligned} \quad (\text{A04OID.45})$$

No coefficient of the conductor or part of the center displacement has been suppressed.

Let $\nu_{r,1}^-$ be the original complete monic lower-root minimum: the minimum of $\|\chi_{k-8} H\|_{\sigma, c-4}^2$ over all monic H of degree r . At $N = q - 1, q$ put $r_N = N - v_A - q' \geq 0$ and let $P = T_N x$ for $x \in K$. If its S^N coefficient is $a_N(P) \neq 0$, the polynomial $\mathcal{C}_A P$ has degree $N - v_A$, is divisible by the full χ_{k-8} , and has leading coefficient $(N)_{v_A} a_N(P)$. Therefore

$$\|x\|_{G_N}^2 = \|P\|_{m_k}^2 \geq \ell_k M_N^{-2} (N)_{v_A} \nu_{r_N,1}^- |a_N(P)|^2. \quad (\text{A04OID.46})$$

When $a_N(P) = 0$ the same inequality is immediate; no nonzero kernel projection is presumed. The original boundary identity (A04OID.5) gives $b_N^* G_N x = \omega_N a_N(P)$. For $K = 0$ the desired kernel bound is immediate. For $K \neq 0$, the supremum of $|b_N^* G_N x|^2 / \|x\|_{G_N}^2$ over nonzero $x \in K$ is $\|P_N b_N\|_{G_N}^2$: Cauchy–Schwarz proves the upper bound and $x = P_N b_N$ attains it when this vector is nonzero, while both sides vanish otherwise. Also $\omega_N \leq u_k \gamma_N$ by applying (A04OID.44) to the monic Gamma minimizer. The exact boundary norms already proved are

$$b_N^* G_N b_N = \omega_N \varepsilon_N^\partial, \quad \varepsilon_{q-1}^\partial = 1, \quad \varepsilon_q^\partial = 1 - d_0.$$

Division only by these positive original factors now proves the source’s conductor inequality, here with its exact receiver:

$$\pi_{1,K,N} \leq \mathcal{B}_N := \min \left\{ 1, \frac{u_k}{\ell_k} \frac{M_N^2 \gamma_N}{\varepsilon_N^\partial (N)_{v_A}^2 \nu_{r_N,1}^-} \right\}, \quad N = q - 1, q. \quad (\text{A04OID.47})$$

It is a bound on the entire actual kernel of the full original observation, with all period dependence still present in its conductor constants.

Combining (A04OID.47) at $N = q$ with (A04OID.41) gives the following new finite allocation of the first source increment:

$$\begin{aligned} d_0 &\leq r_q^B \leq d_0 + (1 - d_0) \mathcal{B}_q, \\ \frac{d_0}{d_0 + (1 - d_0) \mathcal{B}_q} &\leq \kappa_0 \leq 1, \\ 0 &\leq -\log \kappa_0 \leq \log \left(1 + \frac{1 - d_0}{d_0} \mathcal{B}_q \right), \\ -\log(d_0 + (1 - d_0) \mathcal{B}_q) &\leq -\log r_q^B \leq -\log d_0, \\ -\log \kappa_0 - \log r_q^B &= -\log d_0. \end{aligned} \quad (\text{A04OID.48})$$

The lower-root minimum in $\mathcal{B}_q$ can be replaced by its explicit finite scalar lower enclosure in inherited LRS3–4 [7]. More precisely let $\tau_{q',r}$ be that original lower endpoint for $\log \nu_{r,1}^- - \log \gamma_{q'+r}$. Then the same bound holds with $\mathcal{B}_N$ replaced by

$$\widehat{\mathcal{B}}_N = \min \left\{ 1, \frac{u_k}{\ell_k} \frac{M_N^2 \gamma_N}{\varepsilon_N^{\partial}(N)_{v_A}^2 \gamma_{N-v_A}} e^{-\tau_{q',r_N}} \right\}. \quad (\text{A04OID.49})$$

This is a proved finite substitution: the lower endpoint implies $\nu_{r_N,1}^- \geq \gamma_{N-v_A} e^{\tau_{q',r_N}}$, with its original root guards and source mass. It introduces no unknown observation determinant.

For the leading rate, inherited LRS3, LRS11 and LRS14 [7] give on those original finite guards

$$\begin{aligned} \log \nu_{r,1}^- &= \log \gamma_{q'+r} + 2q' \psi(r/q') + O_h(\log q), \\ \psi(t) &= a_0 + \frac{t}{4} \log t + (a_0 - \frac{1}{4})t + O(t^{3/2}), \quad a_0 = \log(4/\pi). \end{aligned} \quad (\text{A04OID.50})$$

These are the inherited complete lower-root norm estimates; they are used for the actual $r_N = N - v_A - q'$, not a changed lower cutoff. Here $q - q' = 16k - 48$, and $r_{q-1} = 16k - 49 - v_A$, $r_q = 16k - 48 - v_A$. Thus

$$\frac{r_N}{q'} = \frac{16}{k} + O(k^{-2}), \quad \frac{r_N}{2} \log \frac{r_N}{q'} = -8k \log k + O(k), \quad \frac{r_N^{3/2}}{\sqrt{q'}} = O(\sqrt{k}).$$

Inserting these three quantities into (A04OID.50) proves

$$2q' \psi(r_N/q') = 2qa_0 - 8k \log k + O_h(k). \quad (\text{A04OID.51})$$

The remaining monic factor has an exact product, retaining every factor from the original derivative:

$$\frac{\gamma_N}{\gamma_{N-v_A}(N)_{v_A}^2} = \prod_{j=0}^{v_A-1} \frac{N-j-1/2}{N-j} \leq 1.$$

For fixed original conductor, (A04OID.45) gives $\log M_N = O_{h,\varpi}(\log q)$. The bound $v_A \leq 80$ is fixed along the original degree family.

For completeness the first-step determinant rate follows from the full physical polynomial rather than an observation inference. Let $R = k\sqrt{\delta^2 + \gamma^2}$ and $\alpha_0 = \pi/2$. The original comparison density has fixed constants $c_G, C_G > 0$ with $c_G e^{-\alpha_0|y|}/\sqrt{1+|y|} \leq \sigma(y) \leq C_G e^{-\alpha_0|y|}/\sqrt{1+|y|}$. For the actual degree- q monic Q , whose full roots lie in $|y| \leq R$, direct integration of $(|y| - R)^q \leq |Q(y)| \leq (|y| + R)^q$ on the lower-tail and full domains respectively gives

$$\mathcal{L}_q(R) \leq \|Q\|_\sigma^2 \leq \mathcal{U}_q(R),$$

where the lower inequality is integrated only over $|y| \geq R+1$ and

$$\begin{aligned} \mathcal{L}_q(R) &= \frac{2c_G e^{-\alpha_0 R}}{\sqrt{R+2}} \alpha_0^{-2q-1/2} \Gamma(2q + \tfrac{1}{2}, \alpha_0), \\ \mathcal{U}_q(R) &= 2C_G e^{\alpha_0 R} \alpha_0^{-2q-1} \Gamma(2q+1). \end{aligned}$$

For the lower bound use $s = |y| - R \geq 1$ and $1 + R + s \leq (R+2)s$; for the upper bound use $s = |y| + R$ and enlarge the positive integral to $s \geq 0$. Monic minimization of (A04OID.44) gives $\ell_k \gamma_q \leq \omega_q \leq u_k \gamma_q$ and hence

$$\frac{\ell_k \gamma_q}{u_k \mathcal{U}_q(R)} \leq d_0 \leq \frac{u_k \gamma_q}{\ell_k \mathcal{L}_q(R)}. \quad (\text{A04OID.52})$$

Stirling's formula and the fixed-shift Gamma-ratio expansion [4, §5.11(i), (iii)] give $\log \mathcal{U}_q(R) - \log \gamma_q = 2q \log(4/\pi) + O_h(k + \log q)$, and the same with $\mathcal{L}_q$. The incomplete Gamma tail changes

its logarithm by $o(1)$: the omitted integral from zero to α_0 is at most $\alpha_0^{2q+1/2}/(2q+1/2)$, while Stirling gives the complete Gamma factor. Therefore

$$\log d_0 = -2qa_0 + O_h(k + \log q), \quad -\log(1 - d_0) = o(1). \quad (\text{A04OID.53})$$

The DLMF expansion is used only for these explicitly displayed Gamma factors; all physical root and source comparison errors remain.

Equations (A04OID.47), (A04OID.44), (A04OID.51) and (A04OID.53) now give the source's actual conductor estimate, with its proof carried into this receiver:

$$\pi_{1,K,N} \leq \mathcal{B}_N \leq \exp\{-2qa_0 + 8k \log k + O_{h,\varpi}(k)\}, \quad N = q - 1, q. \quad (\text{A04OID.54})$$

Finally divide the upper bound at q by the lower estimate for d_0 in (A04OID.53), and insert it in (A04OID.48). This proves the new allocation

$$\begin{aligned} 0 &\leq -\log \frac{\det H_q}{\det H_{q-1}} \leq 8k \log k + O_{h,\varpi}(k), \\ -\log \frac{\det Q_q}{\det Q_{q-1}} &= 2q \log(4/\pi) + O_{h,\varpi}(k \log k), \\ \lim_{k \rightarrow \infty} \frac{-\log(\det H_q / \det H_{q-1})}{q} &= 0, \quad \lim_{k \rightarrow \infty} \frac{-\log(\det Q_q / \det Q_{q-1})}{q} = 2 \log(4/\pi). \end{aligned} \quad (\text{A04OID.55})$$

This evaluates the leading allocation of the single original increment from $q-1$ to q . It neither sums an unproved bound over the other source increments nor assigns either full two-window allocation or an individual projected-current sign. The finite inequality (A04OID.48), or its explicit lower-root enclosure (A04OID.49), remains the receiving formula before passage to the limit.

17 Precisely what has been checked and what is additional

ORE1–5 are the retained definitions, attained observation identity and source increment proved in (A04OID.1)–(A04OID.4) and (A04OID.11)–(A04OID.13). The mathematical contents of ORE6–10 follow from (A04OID.13)–(A04OID.18). The root ORE6 has a typographical error: the literal string `,\quad` must be `,\qquad`. This changes no mathematical factor; the parent integration subsequently corrected that typography. ORE11–17 follow from (A04OID.19)–(A04OID.24), ORE18–19 from (A04OID.31)–(A04OID.32), and ORE20–23 from (A04OID.22) and (A04OID.25)–(A04OID.28). No correction of a sign, conjugation, cutoff, or positive denominator is needed in ORE1–23 on its stated original simple-quartet eigenclass domain.

The supplied low-cutoff note explicitly contains the first and second energy formulas (A04OID.34) and (A04OID.36), the two attained lifts through cutoff q , asymptotic estimates for those energies, a separate conductor bound for the actual kernel overlap, and the one-step class norm ratio in its equation (17). It does not contain the following ORE receiving identities:

1. ORE6–7: the exact restriction/observation determinant allocation and its two logarithmic increments for the same source step.
2. ORE9–12: the quotient numerator update, consecutive transport $U_{N+1} = 1 - (1 - V_N)/d_{K,N}$, and the consecutive response products with their actual earlier conjugation and determinant denominator.
3. ORE14–16: the exact period-dependent observed-energy loss and the associated update of θ_N . The note's full-class energy ratio does not by itself determine this quotient energy.
4. ORE17: the general propagation of $p_{2,N}$ into $p_{1,N+1}$. The note's explicit low-cutoff expressions satisfy its first instance exactly, but do not state this all-cutoff morphism.

5. ORE18–19: the explicit two-window weighted accumulation of the same determinant and energy increments, with the full conductor/native kernel-metric receiver. These identities do not assert new numerical high-cutoff values.
6. ORE20–23: the undivided observed numerators and consecutive observed drops computing the current, its phase, its finite variation bound, and its exact original recurrence/determinant coefficient.

ORE1–3 and ORE8 retain the shared original source, projection and response definitions. ORE4 is a derived finite minimum identity and ORE13 its general full-class energy consequence; neither should be described as an independent evaluation of an observation response. The note’s first low-cutoff energy change is precisely their specialization (A04OID.37) at $q - 1$.

Additional to both the root ORE identities and the note’s equations (5)–(7), this derivation proves the minimum-lift/projector update (A04OID.15), the next-overlap determinant morphism (A04OID.30), the strict original bound in (A04OID.21), the exact next source minimum and norm in (A04OID.37), the low-cutoff full-current transport (A04OID.38)–(A04OID.39), and the explicit observed-current and determinant receivers (A04OID.40)–(A04OID.41), together with the finite and asymptotic single-increment allocations (A04OID.48) and (A04OID.55). The latter use the complete conductor bound proved above and the explicitly cited inherited LRS lower-root estimates. They are a new consequence for the actual observation, not merely the supplied note’s full-class energy calculation. The note’s absolute-action and marked-family calculations are outside this derivation.

Low-cutoff energies with finite arithmetic concentration

This proof evaluates equations (1)–(21) of the supplied low-cutoff continuation [8]. The comparison used on a restricted integral is the actual pointwise upper envelope CAI17, not merely its consequence for whole polynomial norms. The denominator is bounded by the whole-polynomial lower comparison followed by a restricted integral in the Gamma reference measure. Every integral in the original arithmetic source, its mass, and every primary multiplicity are retained.

18 Original source, complete roots and attained lifts

Retain the original stipulated quartet domain of CAI1 and EC1–6 [9, 10]: fixed $0 < \delta < 1/2$, $\gamma > 2$, and complete common order $m_0 \geq 1$, with

$$\begin{aligned} \rho &= \frac{1}{2} + \delta + i\gamma, & h(s) &= \prod_{\zeta \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - \zeta)^{m_0}, & v_h &= 2\xi/h, \\ w_h(y) &= |v_h(1/2 + iy)|^2/(2\pi), & dm_k(y) &= m_k(y) dy = w_h^{*k}(y) dy, & \mu_h &= \int_{\mathbb{R}} w_h(y) dy, \\ k &\equiv 1 \pmod{4}, & k &\geq 9, & c &= k/2, & e &= 1 + k(m_0 - 1), & q &= e(k + 1)^2, \\ \beta_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, & \chi_k(S) &= \prod_{a,b=0}^k (S - \beta_{ab})^e, \end{aligned}$$

$$\lambda = k\rho = c + d + it, \quad d = k\delta > 0, \quad t = k\gamma > 0, \quad z = (\lambda - c)/i = t - id, \quad R = |z| = k\sqrt{\delta^2 + \gamma^2}. \quad (\text{LCS1})$$

The total source mass is μ_h^k by Tonelli. Evenness follows from $\xi(1-s) = \xi(s)$ and $h(1-s) = h(s)$ on the original line, and persists under convolution. The coordinate is always $S = c + iy$. Inner products are conjugate-linear in the first entry.

Put

$$\begin{aligned} Q(y) &= i^{-q} \chi_k(c + iy) = \prod_{a,b=0}^k [y - (2b - k)\gamma + i(2a - k)\delta]^e, \\ \chi_{k,\lambda}(S) &= \chi_k(S)/(S - \lambda)^e, & b_\lambda &= \chi_{k,\lambda}(\lambda)^{-1}, \\ L_\lambda(S) &= b_\lambda \chi_k(S)/(S - \lambda), & v_\lambda &= [L_\lambda] \in E = \mathbb{C}[S]/(\chi_k). \end{aligned} \quad (\text{LCS2})$$

The β_{ab} are distinct before their displayed repetitions, so $b_\lambda \neq 0$. Multiplication gives $(S - \lambda)L_\lambda = b_\lambda \chi_k$. At the λ primary, L_λ has leading nonzero terminal term $(S - \lambda)^{e-1}$, and at every other primary it vanishes to order e . Consequently $v_\lambda \neq 0$ and $Av_\lambda = \lambda v_\lambda$. This retains the power $e - 1$ when $e > 1$; no derivative of χ_k at a repeated root is used.

The complete roots of Q are closed under conjugation by $(a, b) \mapsto (k - a, b)$ and under negation by $(a, b) \mapsto (k - a, k - b)$. Hence Q is real and even. Every root has modulus at most R , all multiplicities are e , and none is real because k is odd and $\delta > 0$. In particular, pairing all roots with their opposites gives the pointwise inequalities

$$|Q(y)|^2 \leq (y^2 + R^2)^q \quad (y \in \mathbb{R}), \quad |Q(y)|^2 \geq (y^2 - R^2)^q \quad (|y| > R). \quad (\text{LCS3})$$

Indeed each of the $q/2$ pairs contributes $y^2 - \zeta^2$; $y^2 - R^2 \leq |y^2 - \zeta^2| \leq y^2 + R^2$ on $|y| > R$, and the upper inequality holds everywhere. Squaring the product proves LCS3.

Let $p_n(S) = i^n Q_n((S - c)/i)$ be the original monic orthogonal polynomials, with Q_n real and $Q_n(-y) = (-1)^n Q_n(y)$, and set $\omega_n = \|p_n\|_{m_k}^2 > 0$, $b_n = [p_n]$. For $N \geq q - 1$ the remainder map and its minimum section are

$$\begin{aligned} J_N : \mathcal{P}_N &\longrightarrow E, & J_N P &= [P], & \ker J_N &= \chi_k \mathcal{P}_{N-q}, \\ \mathcal{S}_N &= H_N^{-1} J_N^* G_N, & G_N &= (J_N H_N^{-1} J_N^*)^{-1}. \end{aligned} \quad (\text{LCS4})$$

Here $\mathcal{P}_j = \mathbb{C}[S]_{\leq j}$, $\mathcal{P}_{-1} = 0$, and H_N is the Gram of $1, S, \dots, S^N$ in the original source. The remainder map is onto, so its displayed Gram is positive definite. Direct multiplication proves $J_N \mathcal{S}_N = 1$ and $\mathcal{S}_N(E) \perp \ker J_N$. For any other lift its difference from $\mathcal{S}_N$ lies in that kernel, and the two squared norms add. This proves both the minimum and the exact isometry from (E, G_N) to the attained source subspace.

Define the literal integrals

$$\begin{aligned} \nu_0 &= \int_{\mathbb{R}} Q(y)^2 m_k(y) dy, & \nu_1 &= \int_{\mathbb{R}} y^2 Q(y)^2 m_k(y) dy, & d\mu_Q(y) &= \nu_0^{-1} Q(y)^2 m_k(y) dy, \\ a_z &= \int_{\mathbb{R}} \frac{d\mu_Q(y)}{y - z}, & m_z &= \int_{\mathbb{R}} \frac{d\mu_Q(y)}{|y - z|^2}, & s_2 &= \int_{\mathbb{R}} y^2 d\mu_Q(y) = \nu_1 / \nu_0, \\ d_0 &= \omega_q / \nu_0, & d_1 &= \omega_{q+1} / \nu_1, & W_N &= [b_N, b_{N+1}, v_\lambda], & F_N &= W_N^* G_N W_N, & E_N &= (F_N)_{33}. \end{aligned} \tag{LCS5}$$

The auxiliary probability in LCS5 expresses ratios of original integrals; ν_0, ν_1, ω_n and E_N are their original full source values throughout.

The monic minimum property gives $0 < d_0, d_1 \leq 1$. In fact both inequalities are strict. For a positive real density the monic orthogonal polynomial of degree n has n simple real roots: otherwise, multiply its real sign-changing factors to obtain a real polynomial of degree below n whose product with the orthogonal polynomial has constant nonzero sign on the real line, contradicting orthogonality. Equality $d_0 = 1$ would make Q that orthogonal polynomial, although all its roots are nonreal. Equality $d_1 = 1$ would do the same for yQ , which still has those nonreal roots. The positive density needed here follows from the original envelope displayed in LCS13 below.

18.1 The two exact low-cutoff minima

At $N = q - 1$ there is no relation in the admitted source. Polynomial division, monicity, parity and orthogonality give

$$\begin{aligned} \mathcal{S}_{q-1} v_\lambda(c + iy) &= b_\lambda i^{q-1} \frac{Q(y)}{y - z}, & E_{q-1} &= |b_\lambda|^2 \nu_0 m_z, \\ \mathcal{S}_{q-1} b_{q-1} &= p_{q-1}, & \mathcal{S}_{q-1} b_q &= p_q - \chi_k, \\ (F_{q-1})_{11} &= \omega_{q-1}, & (F_{q-1})_{13} &= b_\lambda \omega_{q-1}, \\ (F_{q-1})_{22} &= \nu_0 - \omega_q, & (F_{q-1})_{23} &= i b_\lambda \nu_0 a_z. \end{aligned} \tag{LCS6}$$

For example $\langle \chi_k, L_\lambda \rangle = -i b_\lambda \nu_0 a_z$, whereas $\langle p_q, L_\lambda \rangle = 0$ since $\deg L_\lambda = q - 1$; this proves the displayed complex phase of $(F_{q-1})_{23}$. The coefficient of p_{q-1} in L_λ is its leading coefficient b_λ , proving $(F_{q-1})_{13}$.

At $N = q$ the complete relation space is $\mathbb{C}\chi_k$. Projecting onto precisely this line gives

$$\begin{aligned} \mathcal{S}_q v_\lambda(c + iy) &= b_\lambda i^{q-1} Q(y) ((y - z)^{-1} - a_z), \\ E_q &= |b_\lambda|^2 \nu_0 (m_z - |a_z|^2), & \mathcal{S}_q b_q &= p_q - d_0 \chi_k, \\ (F_q)_{11} &= \omega_q (1 - d_0), & (F_q)_{13} &= i b_\lambda \omega_q a_z. \end{aligned} \tag{LCS7}$$

Here $\langle \chi_k, p_q \rangle = \omega_q$ and $\langle \chi_k, L_\lambda \rangle / \nu_0 = -i b_\lambda a_z$. Since q is even, p_{q+1} and $\chi_k(S)(S - c)$ have the same odd parity in $S - c$. Their difference has degree at most $q - 1$ and is orthogonal to χ_k . It is consequently the attained lift at $N = q$ of b_{q+1} . In particular the monic relation $\chi_k(S)(S - c)$ has no missing preceding constant-relation projection. This yields

$$\begin{aligned} \mathcal{S}_q b_{q+1} &= p_{q+1} - \chi_k(S)(S - c), & (F_q)_{22} &= \nu_1 - \omega_{q+1} = \nu_1 (1 - d_1), \\ \int_{\mathbb{R}} \frac{y}{y - z} d\mu_Q(y) &= 1 + z a_z, & (F_q)_{23} &= b_\lambda \nu_0 (1 + z a_z). \end{aligned} \tag{LCS8}$$

The last phase follows from $\overline{i^{q+1}}i^{q-1} = -1$ and the preceding minus sign in the attained lift. The term involving $a_z \int y d\mu_Q$ is zero.

The exact original energies are therefore

$$\boxed{\begin{aligned} p_{1,q-1} &= \frac{\omega_{q-1}}{\nu_0 m_z}, & p_{2,q-1} &= \frac{|a_z|^2}{(1-d_0)m_z}, \\ p_{1,q} &= \frac{d_0|a_z|^2}{(1-d_0)(m_z - |a_z|^2)}, & p_{2,q} &= \frac{|1+za_z|^2}{s_2(1-d_1)(m_z - |a_z|^2)}. \end{aligned}} \quad (\text{LCS9})$$

They equal $|(F_N)_{j3}|^2/[E_N(F_N)_{jj}]$ with exactly the original G_N . The complete-primary factor cancels only between these matching numerator and denominator terms. The variance $m_z - |a_z|^2$ is positive because $(y-z)^{-1}$ is not constant on a positive-density real interval. Also

$$\begin{aligned} \text{Im } a_z &= -dm_z < 0, & 1 + za_z &= \int_{\mathbb{R}} \frac{y^2}{y^2 - z^2} d\mu_Q(y), \\ \text{Im}(1 + za_z) &= -2td \int_{\mathbb{R}} \frac{y^2}{|y^2 - z^2|^2} d\mu_Q(y) < 0. \end{aligned} \quad (\text{LCS10})$$

Thus every displayed response denominator and both response numerators are nonzero.

19 The original arithmetic envelopes and finite norm intervals

The constants in the following display are exactly CAI16–18 and EC14–15 [9, 10], with different subscripts for the Gamma density constants to avoid confusing them with unrelated scalar coefficients:

$$\begin{aligned} \alpha &= \pi/2, \quad p_h = 8m_0 - \frac{9}{2}, \quad B_h = 42 + 8m_0, \quad M_h = 21 + 4m_0, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy > 0, \quad M_\alpha = \int_{\mathbb{R}} e^{\alpha y} w_h(y) dy \in (0, \infty), \\ c_G &= \sqrt{2}e^{-7/6}, \quad C_G = \sqrt{2\sqrt{5}}e^{2/3}, \quad \sigma(y) = |\Gamma(1/4 + iy/2)|^2/(2\pi), \\ \underline{a}_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{C_G 2^{B_h/2}}, \quad \bar{a}_k = \frac{C_h^{\text{tilt}}}{c_G} k^{p_h+1} M_\alpha^{k-1}, \\ D_n &= [1 + 4(n + M_h)^2]^{M_h}, \quad \ell_k = \underline{a}_k / D_{2q}, \quad u_k = \bar{a}_k, \quad \mathcal{L}_k = \log(u_k / \ell_k). \end{aligned} \quad (\text{LCS11})$$

The positive c_h and C_h^{tilt} are the original TW7 and BT12 constants retained in CAI17. Their proved defining envelopes and the Gamma envelopes are

$$\begin{aligned} m_k(y) &\geq c_h \vartheta_h^{k-3} e^{-\alpha(|y|+k-3)} (k-2+|y|)^{-B_h}, \\ m_k(y) &\leq k C_h^{\text{tilt}} M_\alpha^{k-1} e^{-\alpha|y|} (1+|y|/k)^{-p_h}, \\ c_G e^{-\alpha|y|} (1+|y|)^{-1/2} &\leq \sigma(y) \leq C_G e^{-\alpha|y|} (1+|y|)^{-1/2}. \end{aligned} \quad (\text{LCS12})$$

These are established properties of the stipulated arithmetic source, not new premises selected to establish the present result. Since $p_h > 1/2$ and $1 + |y|/k \geq (1 + |y|)/k$, the upper bounds imply

$$0 < \underline{a}_k (1 + y^2)^{-M_h} \sigma(y) \leq m_k(y) \leq u_k \sigma(y) \quad (y \in \mathbb{R}). \quad (\text{LCS13})$$

For the lower inequality use $k - 2 + |y| \leq (k - 2)(1 + |y|)$ and $(1 + |y|)^{B_h} \leq 2^{B_h/2} (1 + y^2)^{M_h}$ in LCS12; the extra factor $(1 + |y|)^{-1/2}$ in the Gamma upper envelope only decreases the proposed lower expression.

The Gamma reference family is Meixner–Pollaczek. Its formulas are those in DLMF Chapter 18, authored by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt

[12]. Equations 18.19.8–9, at the parameters $\lambda_{\text{MP}} = 1/4$, $\varphi = \pi/2$ and $x = y/2$, give the exact transport

$$Q_n^\sigma(y) = n!P_n^{(1/4)}(y/2; \pi/2), \quad \sigma(y) dy = \pi^{-1}w^{(1/4)}(x; \pi/2) dx.$$

In particular Q_n^σ is monic and the transported mass is $\sqrt{2\pi}$. Their generating function, equation 18.23.7, becomes

$$\sum_{n \geq 0} Q_n^\sigma(y) \frac{u^n}{n!} = (1 - iu)^{-1/4+iy/2} (1 + iu)^{-1/4-iy/2} = (1 + u^2)^{-1/4} e^{y \arctan u}, \quad |u| < 1.$$

Differentiation gives $(1 + u^2)\partial_u \mathcal{G} = (y - u/2)\mathcal{G}$; comparison of coefficients proves $Q_{n+1}^\sigma = yQ_n^\sigma - n(n - 1/2)Q_{n-1}^\sigma$, also the specialization of DLMF 18.22.8. DLMF's notes attribute the generating function to M. E. H. Ismail (2009), equation 5.9.3, the recurrence to 5.9.1, and the weight and norms to 5.9.8–9. These are DLMF-reported book ancestry; the underlying book pages were not newly inspected here.

For clarity, the corresponding whole-polynomial lower comparison also follows directly. The transported orthonormal recurrence is

$$ye_j = \sqrt{(j+1)(j+1/2)}e_{j+1} + \sqrt{j(j-1/2)}e_{j-1}.$$

Each of the raising and lowering parts has norm at most $N+1$ on polynomials of degree at most N , so $\|yP\|_\sigma \leq 2(N+1)\|P\|_\sigma$. Iteration and binomial expansion show $\int |P|^2(1+y^2)^{M_h} \sigma \leq D_N \|P\|_\sigma^2$. Cauchy–Schwarz with the two weights $(1+y^2)^{\pm M_h/2}$, then LCS13, gives

$$\ell_k \|P\|_\sigma^2 \leq \|P\|_{m_k}^2 \leq u_k \|P\|_\sigma^2 \quad (\deg P \leq 2q), \quad \mathcal{L}_k = O_h(k + \log(q+1)). \quad (\text{LCS14})$$

The last estimate follows by taking logarithms in LCS11; all h -dependent exponents and constants are fixed and q is a polynomial in k of degree two or three.

For any complex monic polynomial P of degree n with all roots of modulus at most R , define

$$\begin{aligned} \mathcal{L}_n(R) &= \frac{2c_G e^{-\alpha R}}{\sqrt{R+2}} \alpha^{-2n-1/2} \Gamma(2n + \tfrac{1}{2}, \alpha), \\ \mathcal{U}_n(R) &= 2C_G e^{\alpha R} \alpha^{-2n-1} \Gamma(2n+1). \end{aligned} \quad (\text{LCS15})$$

Then

$$\mathcal{L}_n(R) \leq \|P\|_\sigma^2 \leq \mathcal{U}_n(R). \quad (\text{LCS16})$$

For the upper bound use $|P(y)| \leq (|y| + R)^n$, discard the factor $(1 + |y|)^{-1/2} \leq 1$, put $s = |y| + R$, and extend its integration domain to $s \geq 0$. For the lower bound integrate both real half-lines $|y| \geq R+1$, use $|P(y)| \geq (|y| - R)^n$, put $s = |y| - R \geq 1$, and use $1 + R + s \leq (R+2)s$. These operations give exactly LCS15.

The Gamma monic norm in these original mass conventions is

$$\gamma_n = \sqrt{2\pi} n!(1/2)_n = \sqrt{2\pi} \Gamma(2n+1)/4^n. \quad (\text{LCS17})$$

Indeed the monic version of the displayed recurrence has coefficient $n(n - 1/2)$; pairing it against consecutive monic polynomials gives $\gamma_n = n(n - 1/2)\gamma_{n-1}$, starting at the original Gamma mass $\gamma_0 = \sqrt{2\pi}$. The monic minimum and LCS14 imply $\ell_k \gamma_n \leq \omega_n \leq u_k \gamma_n$. Consequently LCS16, applied to the full polynomials $Q/(y-z)$, Q , and yQ , proves all three finite intervals

$$\begin{aligned} \frac{\ell_k}{u_k} \frac{\gamma_n}{\mathcal{U}_n(R)} &\leq x_n \leq \frac{u_k}{\ell_k} \frac{\gamma_n}{\mathcal{L}_n(R)}, \\ (n, x_n) &= (q-1, p_{1,q-1}), (q, d_0), (q+1, d_1). \end{aligned} \quad (\text{LCS18})$$

No root is removed except for the single factor $y - z$ explicitly present in the literal terminal eigenclass.

Here is an elementary control of the logarithmic errors in LCS18. The classical Euler integral and Gamma recurrence are attributed to R. A. Askey and R. Roy's DLMF Chapter 5, equations 5.2.1 and 5.5.1 [13]; the incomplete integrals are explicitly defined below. For $n \geq 1$, shifting the incomplete Gamma integral gives $e^{-\alpha}\Gamma(2n+1/2) \leq \Gamma(2n+1/2, \alpha) \leq \Gamma(2n+1/2)$. Further,

$$\frac{\sqrt{\pi}}{4n+1} \leq \frac{\Gamma(2n+1/2)}{\Gamma(2n+1)} = \sqrt{\pi} \frac{\binom{4n}{2n}}{4^{2n}} \leq \sqrt{\pi}.$$

The inequalities follow because the central binomial coefficient is the largest of the $4n+1$ coefficients whose sum is 4^{2n} . Substitution in LCS15–18, retaining $a_0 = \log(4/\pi) > 0$, proves

$$\begin{aligned} \log p_{1,q-1} &= -2(q-1)a_0 + O_h(k + \log q), \\ \log d_0 &= -2qa_0 + O_h(k + \log q), \\ \log d_1 &= -2(q+1)a_0 + O_h(k + \log q). \end{aligned} \tag{LCS19}$$

In particular d_0, d_1 vanish exponentially. This proof controls the incomplete Gamma term itself and requires no unquantified asymptotic replacement of its lower endpoint.

20 Finite concentration in the original arithmetic measure

Put $Y = Y_q = 2q/\alpha = 4q/\pi$. On the finite guard $Y > R$, LCS3 and the *complete* lower comparison give

$$\begin{aligned} \nu_0 &\geq \ell_k \int_{\mathbb{R}} Q(y)^2 \sigma(y) dy \\ &\geq \ell_k \int_{Y \leq |y| \leq Y+1} Q(y)^2 \sigma(y) dy \geq D_* := \frac{2\ell_k c_G (Y^2 - R^2)^q e^{-\alpha(Y+1)}}{\sqrt{Y+2}}. \end{aligned} \tag{LCS20}$$

Thus D_* does not assert $\int_A Q^2 m_k \geq \ell_k \int_A Q^2 \sigma$ for a restricted set A . The restricted integral in the second line is entirely in the reference measure.

For $0 < b < 1$ write $A_b = \{y : (1-b)Y \leq |y| \leq (1+b)Y\}$, and for $j = 0, 2$ define

$$\begin{aligned} U_j(b) &= 2u_k C_G \sum_{r=0}^q \binom{q}{r} R^{2(q-r)} \alpha^{-(2r+j+1/2)} [\gamma(2r+j+\tfrac{1}{2}, \alpha(1-b)Y) \\ &\quad + \Gamma(2r+j+\tfrac{1}{2}, \alpha(1+b)Y)]. \end{aligned} \tag{LCS21}$$

Here $\gamma(s, x) = \int_0^x u^{s-1} e^{-u} du$ and $\Gamma(s, x) = \int_x^\infty u^{s-1} e^{-u} du$. The pointwise upper inequality in LCS13 is valid on each part of A_b^c . Apply it there, use LCS3, expand $(y^2 + R^2)^q$, and use $(1+y)^{-1/2} \leq y^{-1/2}$ for $y > 0$. Every coefficient is nonnegative, all endpoint integrals are finite, and direct integration proves

$$\int_{A_b^c} |y|^j Q(y)^2 m_k(y) dy \leq U_j(b), \quad \int_{A_b^c} |y/Y|^j d\mu_Q(y) \leq \frac{U_j(b)}{Y^j D_*}. \tag{LCS22}$$

This proves the literal finite enclosure (13) of [8] from an actual density inequality.

20.1 A proved exponential bound for the finite tails

To evaluate the tails without an unsupported Laplace assertion, define $n_j = 2q + j$, $a_j = n_j + 1$, and

$$\begin{aligned} \tilde{U}_j(b) &= 2u_k C_G e^{\alpha R} \alpha^{-a_j} [\gamma(a_j, \alpha((1-b)Y + R)) \\ &\quad + \Gamma(a_j, \alpha((1+b)Y + R))]. \end{aligned} \tag{LCS23}$$

This is another upper bound for the left side of LCS22. Indeed, for $y \geq 0$, $y^j (y^2 + R^2)^q (1+y)^{-1/2} \leq (y+R)^{2q+j}$. The substitution $s = y + R$ maps the low interval into $[0, (1-b)Y + R]$ after a positive extension, and maps the high interval onto $[(1+b)Y + R, \infty)$. This gives LCS23 exactly.

For $a > 0$ the density $u^{a-1}e^{-u}/\Gamma(a)$ has exponential moment $(1-\tau)^{-a}$ for $\tau < 1$, by the substitution $(1-\tau)u = v$ in its defining integral. Markov's inequality with $\tau = 1 - r^{-1}$, which is negative for $0 < r < 1$ and positive for $r > 1$, therefore proves

$$\frac{\gamma(a, ar)}{\Gamma(a)} \leq e^{-ag(r)} \quad (0 < r < 1), \quad \frac{\Gamma(a, ar)}{\Gamma(a)} \leq e^{-ag(r)} \quad (r > 1), \quad g(r) = r - 1 - \log r. \quad (\text{LCS24})$$

For the lower-tail case the event $u \leq ar$ implies $e^{\tau u} \geq e^{\tau ar}$ because $\tau < 0$, which justifies the same Markov argument and its sign. The function g is positive away from $r = 1$, as follows from $g'(r) = 1 - r^{-1}$.

Set

$$\begin{aligned} r_{j,-} &= \frac{2q(1-b) + \alpha R}{2q + j + 1}, & r_{j,+} &= \frac{2q(1+b) + \alpha R}{2q + j + 1}, \\ C_{q,j} &= \frac{2u_k C_G e^{\alpha R} \alpha^{-a_j} \Gamma(a_j)}{Y^j D_*}, & \mathcal{E}_j(b) &= C_{q,j} \left(e^{-a_j g(r_{j,-})} + e^{-a_j g(r_{j,+})} \right). \end{aligned} \quad (\text{LCS25})$$

On the explicit finite guards $Y > R$ and $0 < r_{j,-} < 1 < r_{j,+}$, LCS23–24 give

$$\int_{A_b^c} |y/Y|^j d\mu_Q(y) \leq \frac{\tilde{U}_j(b)}{Y^j D_*} \leq \mathcal{E}_j(b). \quad (\text{LCS26})$$

Every prefactor here can be evaluated from the original constants. Its size has the following explicit bound:

$$\begin{aligned} \log C_{q,j} &\leq \mathcal{L}_k + \log(C_G/c_G) + \alpha R + \alpha + \frac{1}{2} \log(Y+2) - \log \alpha \\ &\quad + 1 + \log(2q+j) + \frac{j^2}{2q} - q \log(1 - R^2/Y^2). \end{aligned} \quad (\text{LCS27})$$

To verify it, substitution of D_* gives the exact expression

$$C_{q,j} = \frac{u_k C_G}{\ell_k c_G} \frac{e^{\alpha R + 2q + \alpha} \sqrt{Y+2}}{\alpha} \frac{(2q+j)!}{(2q)^{2q+j}} (1 - R^2/Y^2)^{-q}.$$

For an integer $n \geq 1$, monotonicity of $\log x$ on the intervals $[r, r+1]$ implies $\log(n!) \leq n \log n - n + 1 + \log n$. Taking $n = 2q + j$ and using $\log(1 + j/(2q)) \leq j/(2q)$ proves LCS27.

Now $R = O_h(k)$, $q \geq (k+1)^2$ and $R/Y = O_h(k/q) \rightarrow 0$. On the eventual guard $R^2/Y^2 \leq 1/2$, $-q \log(1 - R^2/Y^2) \leq 2qR^2/Y^2 = O_h(k^2/q)$. Thus $\log C_{q,j} = O_h(k + \log q) = o_h(q)$. For every fixed $b \in (0, 1)$, $r_{j,-} \rightarrow 1 - b$ and $r_{j,+} \rightarrow 1 + b$, so $\min\{g(r_{j,-}), g(r_{j,+})\}$ has a strictly positive limit. LCS26 consequently proves, for each such fixed b ,

$$\int_{A_b^c} d\mu_Q \rightarrow 0, \quad \int_{A_b^c} (y/Y)^2 d\mu_Q \rightarrow 0 \quad \text{exponentially in } q. \quad (\text{LCS28})$$

The literal sum in LCS21 has the same exponential conclusion. Indeed its integral form is

$$2u_k C_G \int_{[0, (1-b)Y] \cup [(1+b)Y, \infty)} y^{j-1/2} (y^2 + R^2)^q e^{-\alpha y} dy.$$

On $y \geq 1$ its integrand is bounded by the shifted integrand used in LCS23; on $0 < y < 1$ bound $(y^2 + R^2)^q$ by $(1 + R^2)^q$ and integrate $y^{j-1/2}$. Thus the additional finite bound is

$$\frac{U_j(b)}{Y^j D_*} \leq \mathcal{E}_j(b) + \frac{u_k C_G}{\ell_k c_G} \frac{e^{2q + \alpha} \sqrt{Y+2}}{(j+1/2)Y^j} \left(\frac{1 + R^2}{Y^2 - R^2} \right)^q. \quad (\text{LCS28a})$$

The logarithm of the last term is at most $O_h(k + \log q) + 2q + q \log((1 + R^2)/(Y^2 - R^2))$. Its logarithm divided by q tends to $-\infty$ because $(1 + R^2)/(Y^2 - R^2) = O_h(k^2/q^2) \rightarrow 0$. Consequently both $U_0(b)/D_*$ and $U_2(b)/(Y^2 D_*)$ themselves vanish exponentially, as asserted in the supplied continuation. This conclusion uses the complete root polynomial and the actual arithmetic upper envelope. LCS26 also supplies an independent finite bound if the size of the sum in LCS21 makes it inconvenient.

21 The three scalar integrals and evaluated energies

One may take the finite bounds

$$\eta_b = \min\{1, U_0(b)/D_*, \mathcal{E}_0(b)\}, \quad \tau_b = \min\{U_2(b)/(Y^2 D_*), \mathcal{E}_2(b)\}$$

when the LCS25 guards hold; omitting $\mathcal{E}_j$ defines valid bounds without those extra guards. Evenness gives $a_z = z \int (y^2 - z^2)^{-1} d\mu_Q(y)$. For every real y , $|y - z| \geq d$ and $|y^2 - z^2| \geq 2td$. On the annulus A_b , triangle inequalities now give, whenever $(1 - b)Y > R$,

$$\begin{aligned} \frac{1 - \eta_b}{((1 + b)Y + R)^2} &\leq m_z \leq \frac{1}{((1 - b)Y - R)^2} + \frac{\eta_b}{d^2}, \\ \left| \frac{Y^2 a_z}{z} - 1 \right| &\leq B_b := \frac{2b + b^2 + R^2/Y^2}{(1 - b)^2 - R^2/Y^2} + \eta_b \left(1 + \frac{Y^2}{2td} \right), \\ (1 - b)^2(1 - \eta_b) &\leq \frac{s_2}{Y^2} \leq (1 + b)^2 + \tau_b. \end{aligned} \tag{LCS29}$$

For the middle estimate use

$$\left| \frac{Y^2}{y^2 - z^2} - 1 \right| \leq \frac{2b + b^2 + R^2/Y^2}{(1 - b)^2 - R^2/Y^2} \quad (y \in A_b),$$

and bound the same integrand by $1 + Y^2/(2td)$ on A_b^c . For the last estimate retain the second-moment contribution τ_b on A_b^c rather than discarding the tail.

The quantities Y^2/d^2 and $Y^2/(td)$ grow at most polynomially in k , since $d = k\delta$, $t = k\gamma$, and q has degree two or three in k . For fixed b , their products with η_b therefore vanish by LCS28. First take $k \rightarrow \infty$ in LCS29 and then let $b \downarrow 0$. This proves all three limits

$$\boxed{Y^2 m_z \rightarrow 1, \quad Y^2 a_z/z \rightarrow 1, \quad s_2/Y^2 \rightarrow 1.} \tag{LCS30}$$

The exponentially small central mass has been controlled even after multiplication by the inverse denominators.

For explicit finite energy intervals, write $r_0 = R^2/Y^2$ and

$$\begin{aligned} M_- &= \frac{1 - \eta_b}{(1 + b + R/Y)^2}, \quad M_+ = \frac{1}{(1 - b - R/Y)^2} + \frac{Y^2 \eta_b}{d^2}, \\ A_- &= (1 - B_b)_+, \quad A_+ = 1 + B_b, \quad S_- = (1 - b)^2(1 - \eta_b), \quad S_+ = (1 + b)^2 + \tau_b, \\ H_- &= M_- - r_0 A_+^2, \quad H_+ = M_+ - r_0 A_-^2. \end{aligned} \tag{LCS31}$$

On the displayed finite guards and $\eta_b < 1$, $H_- > 0$, LCS9 and LCS29 give

$$\begin{aligned} \frac{r_0 A_-^2}{(1 - d_0)M_+} &\leq p_{2,q-1} \leq \frac{r_0 A_+^2}{(1 - d_0)M_-}, \\ \frac{d_0 r_0 A_-^2}{(1 - d_0)H_+} &\leq p_{1,q} \leq \frac{d_0 r_0 A_+^2}{(1 - d_0)H_-}, \\ \frac{(1 - r_0 A_+)_+^2}{(1 - d_1)S_+ H_+} &\leq p_{2,q} \leq \frac{(1 + r_0 A_+)^2}{(1 - d_1)S_- H_-}. \end{aligned} \tag{LCS32}$$

The guards hold for every sufficiently large k at a fixed sufficiently small $b > 0$, as follows from LCS28 and $r_0 \rightarrow 0$. For the last line use $|za_z| \leq r_0 A_+$. The inputs d_0, d_1 themselves have the explicit finite intervals LCS18. Thus the finite enclosures retain actual source integrals and also admit enclosure solely by the original comparison constants.

Substituting LCS30 and LCS19 into the exact LCS9 gives

$$\boxed{\begin{aligned} p_{1,N} &= \exp[-2q \log(4/\pi) + O_h(k + \log q)], \quad N = q - 1, q, \\ p_{2,q-1} &= \frac{\pi^2(\delta^2 + \gamma^2)}{16} \frac{k^2}{q^2} (1 + o_h(1)), \\ p_{2,q} &= 1 - o_h(1), \\ p_{1,q} &= d_0 \frac{\pi^2(\delta^2 + \gamma^2)}{16} \frac{k^2}{q^2} (1 + o_h(1)). \end{aligned}} \quad (\text{LCS33})$$

To see the variance step explicitly, $Y^2|a_z|^2 = (R^2/Y^2)|Y^2 a_z/z|^2 \rightarrow 0$, so $Y^2(m_z - |a_z|^2) \rightarrow 1$. Also $z a_z \rightarrow 0$ and $s_2/Y^2 \rightarrow 1$, giving $p_{2,q} \rightarrow 1$. The one-sided notation in LCS33 is justified by the exact projection inequality $p_{2,q} \leq 1$, proved below. The logarithm of R^2/Y^2 is $O_h(\log q)$, which explains the stated exponential rate for $p_{1,q}$. Finally the same-class norm ratio is exactly, and then asymptotically,

$$\boxed{\frac{E_q}{E_{q-1}} = 1 - \frac{|a_z|^2}{m_z} = 1 - \frac{\pi^2(\delta^2 + \gamma^2)}{16} \frac{k^2}{q^2} (1 + o_h(1)).} \quad (\text{LCS34})$$

22 The original response matrix, spectrum and second response

Reflection $P(S) \mapsto P(k - S)$ is an isometry in the even arithmetic source. It preserves $\chi_k \mathcal{P}_{N-q}$ and therefore commutes with the attained minimum section LCS4. The two source polynomials p_N, p_{N+1} have opposite parities, as do their attained lifts. Consequently $(F_N)_{12} = 0$. Their normalized quotient classes are orthonormal, and projection of $v_\lambda/\sqrt{E_N}$ onto their span proves $0 < p_{1,N}, p_{2,N}$ and $p_{1,N} + p_{2,N} \leq 1$.

For completeness, retain the exact JRE map [11]. Denote the original quotient by $E_{\text{packet}} = \mathbb{C}[S]/(\chi_k)$ to distinguish it from the following scalar. At one of the two cutoffs write $v = v_\lambda$, $E = E_N$, $f = (F_{13}, F_{23})^\top$, $B = [b_N, b_{N+1}]$, and

$$\begin{aligned} A &= B - v f^*/E : \mathbb{C}^2 \longrightarrow (E_{\text{packet}}, G_N), \\ S &= A^* G_N A, \quad D = \text{diag}(F_{13}, F_{23}), \quad T = D^{-1} S D^{-*}. \end{aligned} \quad (\text{LCS35})$$

Expanding $S = F_{[1,2],[1,2]} - f f^*/E$ and using $F_{12} = 0$ proves the exact equality with its original source factor

$$\boxed{ET = \begin{pmatrix} p_1^{-1} - 1 & -1 \\ -1 & p_2^{-1} - 1 \end{pmatrix}.} \quad (\text{LCS36})$$

The eigenvalues of T are those in the next display divided by the original E ; its mass has not been replaced.

The exact eigenvalues of ET are

$$\lambda_{\pm} = \frac{1}{2} \left[p_1^{-1} + p_2^{-1} - 2 \pm \sqrt{(p_1^{-1} - p_2^{-1})^2 + 4} \right]. \quad (\text{LCS37})$$

At $N = q - 1$, put $a = p_1^{-1} - 1$, $b' = p_2^{-1} - 1$. LCS33 implies $a/b' \rightarrow \infty$ and $b' \rightarrow \infty$. For $a > b'$ the exact deviation is

$$\lambda_+ = a + \Delta, \quad \lambda_- = b' - \Delta, \quad \Delta = \frac{2}{\sqrt{(a - b')^2 + 4} + (a - b')} \in (0, (a - b')^{-1}].$$

It follows that $\lambda_+/a \rightarrow 1$ and $\lambda_-/b' \rightarrow 1$, in particular that ET is positive definite eventually. Thus

$$\boxed{\begin{aligned} \lambda_{\max}(E_{q-1} T_{q-1}) &= \exp[2q \log(4/\pi) + O_h(k + \log q)], \\ \lambda_{\min}(E_{q-1} T_{q-1}) &= \frac{16}{\pi^2(\delta^2 + \gamma^2)} \frac{q^2}{k^2} (1 + o_h(1)), \\ \lim_{k \rightarrow \infty} \frac{1}{q} \log \text{cond}(E_{q-1} T_{q-1}) &= 2 \log(4/\pi). \end{aligned}} \quad (\text{LCS38})$$

Finally let P_K be any of the admitted G_N -orthogonal observation-kernel projections. Retain $K = W_N^* G_N P_K W_N$ with the three columns (b_N, b_{N+1}, v) , and define $\theta = K_{33}/E \in [0, 1]$, $V = K_{23}/F_{23}$. The vector $x = P_K v - \theta v$ satisfies

$$\langle v, x \rangle_{G_N} = 0, \quad \|x\|_{G_N}^2 = E\theta(1 - \theta).$$

The residual second column $r = b_{N+1} - v\overline{F_{23}}/E$ has norm squared $F_{22} - |F_{23}|^2/E$. Applying Cauchy–Schwarz to $\langle r, x \rangle = K_{23} - \theta F_{23}$ gives the full finite bound

$$\boxed{|V - \theta|^2 \leq \theta(1 - \theta)(p_2^{-1} - 1)}. \quad (\text{LCS39})$$

At $N = q$, LCS33 implies

$$|V_q - \theta_q| \leq \sqrt{\theta_q(1 - \theta_q)(p_{2,q}^{-1} - 1)} = o_h(1), \quad 0 \leq \theta_q(1 - \theta_q) \leq \frac{1}{4}. \quad (\text{LCS40})$$

The convergence is uniform over those projections. It proves $\text{Im } V_q = o_h(1)$ and $\text{Re } V_q - \theta_q = o_h(1)$ without assigning a value to θ_q . For the actual joint product the exact residual is $\overline{U}_q(V_q - \theta_q)$, whose modulus is $|U_q| |V_q - \theta_q|$. No unproved bound on U_q is used in deducing LCS40, and that factor cannot be omitted when passing to the product.

Both parity subsequences and the terminal cutoff

Abstract

The two boundary projection energies in the original response ellipse are evaluated through the full retained eigenclass, its consecutive Cauchy transforms, and its actual source-norm drops. The eigen-equation determines the complete opposite-parity masses. Their exact transport between adjacent attained sections expresses each class-norm drop in the original directed phase errors. Together with the original arithmetic covariance and multiplication estimates, this proves $p_2 = \Theta_h(a^2/q^2)$ on the even cutoff subsequence and $p_1 = O_h(q^{-2})$ on the odd subsequence, with explicit finite bounds and complete primary multiplicity retained. The same directed error extends to the previously uncovered final receiving cutoff. The observation-kernel projection remains its original map; none of its individual projected signs is inferred from these energies.

23 Original packet, masses and minimum

This continuation starts from the complete boundary and current maps NCE1–21 [14], the original response ellipse JRE1–22 [15], and the actual minimum and rank-two formulas PT21–26 [16]. The earlier product identity NCE21 is retained as an input already established there. The new results below evaluate the individual factors through adjacent minimum sections and their complete parity energies. The source-specific analytic estimates used are precisely CAI16–28 [17] and EC15–35 [18].

Use a for the original tensor degree, with $a \equiv 1 \pmod{4}$, $a \geq 5$, and retain

$$\begin{aligned} \rho &= \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m_0 \geq 1, \quad h(s) = \prod_{\zeta \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - \zeta)^{m_0}, \\ w_h(y) &= |(2\xi/h)(1/2 + iy)|^2/(2\pi), \quad dm_a(y) = w_h^{*a}(y) dy, \quad c = a/2, \\ e &= 1 + a(m_0 - 1), \quad q = e(a + 1)^2, \quad \chi(S) = \prod_{u,v=0}^a [S - c - (2u - a)\delta - i(2v - a)\gamma]^e, \quad (\text{RBE1}) \\ E &= \mathbb{C}[S]/(\chi), \quad A = M_S, \quad \lambda = a\rho = c + d + it, \quad d = a\delta > 0, \quad t = a\gamma > 0, \\ H_\lambda &= \chi_\lambda(\lambda), \quad \chi_\lambda(S) = \chi(S)/(S - \lambda)^e, \quad v_\lambda = [L_\lambda], \quad L_\lambda(S) = \chi(S)/[(S - \lambda)H_\lambda]. \end{aligned}$$

The factor $H_\lambda \neq 0$ is the full product of the other root differences with multiplicity e . The class is a nonzero terminal primary vector and $Av_\lambda = \lambda v_\lambda$, because $(S - \lambda)L_\lambda = \chi/H_\lambda$. For $e > 1$, the factor $(S - \lambda)^{e-1}$ remains in the class. Its actual arithmetic inclusion retains the factor $[a(m_0 - 1)!]/[(m_0 - 1)!]^a$, the full unit $v_h(\rho)^a$, and all tensor primary terminal powers, as in EC5–6. These factors are not reset when ratios of actual norms are formed.

For $N \geq q - 1$ put

$$\begin{aligned} \mathcal{P}_N &= \mathbb{C}[S]_{\leq N}, \quad S = c + iy, \quad \langle f, g \rangle_a = \int \overline{f(c + iy)} g(c + iy) dm_a(y), \\ \mathcal{V}_N &= \mathcal{P}_N \ominus \chi \mathcal{P}_{N-q}, \quad J_N P = [P]_\chi, \quad G_N = (J_N H_N^{-1} J_N^*)^{-1}, \quad T_N = H_N^{-1} J_N^* G_N, \quad (\text{RBE2}) \\ f_N &= T_N v_\lambda, \quad E_N = \|f_N\|_a^2 > 0. \end{aligned}$$

Here H_N is the original Gram of $1, S, \dots, S^N$. The symbol f_N is the retained minimum polynomial in this paper; boundary unit vectors below have different letters. The minimum statement follows exactly from $J_N T_N = I$ and $T_N \perp \ker J_N$, obtained by matrix multiplication in (RBE2). Every representative is the attained one plus an orthogonal original relation. All moments and all source masses in these expressions are those of dm_a .

Let p_n be the original S -monic orthogonal polynomial, $\omega_n = \|p_n\|_a^2$, and $b_n = [p_n]$. For the three original columns $(b_N, b_{N+1}, v_\lambda)$, write

$$F_{ij} = \langle T_N W_i, T_N W_j \rangle_a, \quad p_{1,N} = \frac{|F_{13}|^2}{E_N F_{11}}, \quad p_{2,N} = \frac{|F_{23}|^2}{E_N F_{22}}, \quad \mathfrak{c}_N = \frac{\overline{F_{13}} F_{23}}{\omega_N E_N}, \quad \kappa_N = \frac{\sqrt{F_{11} F_{22}}}{\omega_N}. \quad (\text{RBE3})$$

These are exactly JRE20 and NCE9,18, with the cutoff now explicit. No observation projection is used in defining $p_{j,N}$.

24 The full eigen-equation evaluates the parity masses

The original source is even and $\chi(a - S) = (-1)^q \chi(S)$; here q is a multiple of four. Thus $\mathcal{R}P(S) = P(a - S)$ is a source isometry preserving every relation space and $\mathcal{V}_N$. The original monic minimum has parity $(-1)^n$, since $(-1)^n p_n(a - S)$ is another monic orthogonal polynomial with the same degree and hence equals p_n . The two vectors $\hat{b}_1 = T_N b_N$ and $\hat{b}_2 = T_N b_{N+1}$ therefore have opposite parities and are orthogonal.

Set $J = \Pi_{\mathcal{V}_N} M_y|_{\mathcal{V}_N}$, which is Hermitian and satisfies $\mathcal{R}J\mathcal{R} = -J$. NCE2–6 give the exact action in this actual source space:

$$T_N A T_N^{-1} = cI + iJ + \frac{\hat{b}_2 \hat{b}_1^*}{\omega_N}. \quad (\text{RBE4})$$

For clarity, its coefficient can be verified without any spectral assumption. The leading coefficient of $T_N x$ is $\omega_N^{-1} \langle p_N, T_N x \rangle = \omega_N^{-1} b_N^* G_N x$. Multiply $T_N x$ by S , subtract this coefficient times p_{N+1} , and project onto $\mathcal{V}_N$. The subtracted remainder is b_{N+1} , so adding its attained representative proves (RBE4), including the positive sign. Integration on $S = c + iy$ gives

$$\Re c_N = d, \quad \mathfrak{c}_N = d + i(\mu_N - t), \quad \mu_N = \frac{\langle f_N, J f_N \rangle_a}{E_N} = \frac{\int y |f_N(c + iy)|^2 dm_a(y)}{E_N}. \quad (\text{RBE5})$$

Indeed pair the eigen-equation from (RBE4) with f_N . Its rank-one contribution is $\overline{\mathfrak{c}_N} E_N$. Equating real and imaginary parts gives exactly (RBE5). In particular F_{13} and F_{23} are nonzero, and both boundary norms are positive.

Define the explicit isometric coordinate vectors $e_N = \hat{b}_1 / \sqrt{F_{11}}$, $g_N = \hat{b}_2 / \sqrt{F_{22}}$, and $x_N = f_N / \sqrt{E_N}$. The original vectors and their masses remain the numerators of these specified coordinate maps. Let P_e, P_g be the two complete parity projections in $\mathcal{V}_N$, with $e_N \in \text{im } P_e$ and $g_N \in \text{im } P_g$, and put

$$u_N = \|P_e x_N\|^2, \quad v_N = \|P_g x_N\|^2, \quad u_N + v_N = 1, \quad \mathfrak{c}_N = d + it(u_N - v_N), \quad \mu_N = 2tu_N. \quad (\text{RBE6})$$

Here is a full proof of the new equality. Write $x = x_e + x_g$ and $\alpha = e_N^* x$, $\beta = g_N^* x$. The eigen-equation is $(d + it)x = iJx + \kappa_N g_N \alpha$. Its first parity component has no rank-one term, so $(d + it)u_N = i\langle x_e, Jx_g \rangle$. The second gives $(d + it)v_N = i\langle x_g, Jx_e \rangle + \kappa_N \beta \alpha$. Hermiticity of J and conjugation of the first equality now give $\overline{\mathfrak{c}_N} = (d + it)v_N + (d - it)u_N$, which proves (RBE6). Both u_N, v_N are strictly positive: if either were zero, one of α, β would vanish, contradicting $\Re c_N = d > 0$. Therefore

$$0 < \mu_N < 2t, \quad p_{1,N} \leq u_N, \quad p_{2,N} \leq v_N, \quad p_{1,N} p_{2,N} = \frac{d^2 + t^2(u_N - v_N)^2}{\kappa_N^2}. \quad (\text{RBE7})$$

The inequalities are the actual inclusion of a boundary line in its whole parity space. The product is NCE21 now expressed through the evaluated complete parity masses.

25 A source multiplication bound on each individual energy

Let $\mathcal{C}_N = C_{\text{mult}}(h)(2N + \lfloor a/3 \rfloor)$ be the original arithmetic multiplication constant, CAI22–25. Then

$$\|J\| \leq \mathcal{C}_N, \quad \kappa_N^2 p_{1,N} = \|(d + it - iJ)x_N\|^2 = |\mathfrak{c}_N|^2 + \text{Var}_{x_N}(J), \quad (\text{RBE8})$$

where $\text{Var}_x(J) = \|Jx\|^2 - \langle x, Jx \rangle^2$. The first inequality is compression of the actual multiplication estimate on degree N , not substitution of a Gamma metric. The second follows by squaring the

eigen-equation; its real scalar d has zero cross term with $i(t - J)$. Subtracting and adding $(t - \mu_N)^2$ proves the last equality using (RBE5). In particular, with $D_N^2 = d^2 + (t + \mathcal{C}_N)^2$,

$$\frac{|\mathbf{c}_N|^2}{\kappa_N^2} \leq p_{1,N} \leq \frac{D_N^2}{\kappa_N^2}, \quad p_{2,N} = \frac{|\mathbf{c}_N|^2}{|\mathbf{c}_N|^2 + \text{Var}_{x_N}(J)} \geq \frac{|\mathbf{c}_N|^2}{D_N^2}. \quad (\text{RBE9})$$

The upper bound in (RBE8) uses $\|t - J\| \leq t + \mathcal{C}_N$. The formula for p_2 follows by dividing the exact product in (RBE7) by (RBE8), with all denominators strictly positive. The first parity equation also gives

$$(d^2 + t^2)u_N = \|P_e J P_g x_N\|^2 \leq \mathcal{C}_N^2 v_N, \quad v_N \geq \frac{d^2 + t^2}{\mathcal{C}_N^2 + d^2 + t^2}. \quad (\text{RBE10})$$

This is an additional finite lower bound on the complete g_N -parity mass, and does not identify that mass with its single boundary-line energy p_2 .

The constant retains the original source masses. To specify it fully, put $M = 21 + 4m_0$, $\alpha_0 = \pi/2$, $M_\alpha = \int e^{\alpha_0 y} w_h(y) dy$, $\vartheta_h = \int_{-1}^1 w_h(y) dy$, and retain the original lower-envelope constant c_h from TW7. With $C_\Gamma^{\text{env}} = \sqrt{2\sqrt{5}e^{2/3}}$, set

$$a_b = \frac{c_h \vartheta_h^{b-3} e^{-\alpha_0(b-3)} (b-2)^{-42-8m_0}}{C_\Gamma^{\text{env}} 2^{(42+8m_0)/2}}, \quad C_\sigma = \frac{4}{\sqrt{15}\sqrt{2\pi}}, \quad d_0 = \pi/2 - \log(5/3) > 0, \\ H_h = \frac{64C_\sigma(1+4M^2)^M}{d_0^2} \max_{b=3,4,5} \frac{M_\alpha^b}{a_b}, \quad (\text{RBE11}) \\ T_h = \frac{2}{d_0} \{\log 16 + 2M + 3 + \log^+ H_h\}, \quad C_{\text{mult}}(h) = \sqrt{T_h^2 + 1}.$$

CAI21 proves the original tail inequality $\int_{|y|>R} y^2 |P|^2 dm_b \leq H_h(N+1)^{2M+2} 16^N e^{-d_0 R/2} \|P\|_{m_b}^2$ for $b = 3, 4, 5$, $\deg P \leq N$. At $R = T_h(N+1)$ the coefficient is at most one, while on the central interval $y^2 \leq T_h^2(N+1)^2$. Thus the full multiplication bound is $C_{\text{mult}}(N+1)$. To pass to a , partition it into $\lfloor a/3 \rfloor$ blocks of sizes 3, 4, 5, and use the exact convolution isometry $P(y) \mapsto P(\sum y_i)$. The total-degree orthonormal product recurrence has absolute row sum at most $C_{\text{mult}}(2N + \lfloor a/3 \rfloor)$, including the entire output degree $N+1$. Symmetry gives this operator bound, proving (RBE8). This restates the receiving proof of CAI25; the constants and its earlier original-source tail estimate are unchanged, and both integration regions contribute.

26 Exact individual Cauchy-transform formulas

Let $r = N + 1 - q$. For the full relation measure $dW(y) = |\chi(c + iy)|^2 dm_a(y)$, let $\pi_j(S)$ be the S -monic orthogonal polynomials and $\nu_j = \int |\pi_j(c + iy)|^2 dW(y)$. Define the actual factors

$$C_j(\lambda) = \int \frac{\overline{\pi_j(c + iy)}}{c + iy - \lambda} dW(y), \quad Z_\lambda = \int \frac{dW(y)}{|c + iy - \lambda|^2}, \quad A_r = Z_\lambda - \sum_{j=0}^{r-1} \frac{|C_j(\lambda)|^2}{\nu_j} > 0. \quad (\text{RBE12})$$

The empty sum at $r = 0$ is zero. Orthogonal projection onto all old relations gives the full polynomial

$$f_N(S) = \frac{\chi(S)}{H_\lambda} \left\{ \frac{1}{S - \lambda} - \sum_{j=0}^{r-1} \frac{C_j(\lambda)}{\nu_j} \pi_j(S) \right\}, \quad E_N = A_r / |H_\lambda|^2. \quad (\text{RBE13})$$

Each term is either the original terminal polynomial or an original χ -multiple. The orthogonality follows directly from the definition of C_j ; positivity of A_r is the strict positivity of the nonzero retained class's minimum norm. This also proves the exact source-norm drop

$$f_{N+1} = f_N - \frac{C_r(\lambda)}{H_\lambda \nu_r} \chi \pi_r, \quad A_{r+1} = A_r - \frac{|C_r(\lambda)|^2}{\nu_r}. \quad (\text{RBE14})$$

For $r \geq 1$, orthogonality of the original monic polynomials and the leading coefficients give

$$\begin{aligned} T_N b_N &= p_N - \frac{\omega_N}{\nu_{r-1}} \chi \pi_{r-1}, & T_N b_{N+1} &= p_{N+1} - \chi \pi_r, \\ F_{11} &= \omega_N (1 - \omega_N / \nu_{r-1}), & F_{22} &= \nu_r - \omega_{N+1}, \\ F_{13} &= -\frac{\omega_N C_{r-1}}{H_\lambda \nu_{r-1}}, & F_{23} &= -\frac{C_r}{H_\lambda}. \end{aligned} \quad (\text{RBE15})$$

To verify the first identity, pair p_N with the old relation basis. Every earlier relation has degree below N , giving zero pairing. The final one $\chi \pi_{r-1}$ is monic of degree N , giving pairing ω_N . Subtraction is therefore the entire orthogonal relation projection. For the second identity the top monic coefficient cancels, and both polynomials are orthogonal to all old relations; the remainder is exactly b_{N+1} . Taking norms and pairing with (RBE13) proves the other four identities. At $r = 0$, replace the first, third and fifth entries by $T_N b_N = p_{q-1}$, $F_{11} = \omega_{q-1}$, and $F_{13} = \omega_{q-1}/H_\lambda$; the remaining entries are unchanged.

Consequently the individual boundary energies, including all original norms and complete root factors before their cancellation, are evaluated exactly by

$$\begin{aligned} p_{1,N} &= \frac{\omega_N |C_{r-1}|^2}{A_r \nu_{r-1} (\nu_{r-1} - \omega_N)} \quad (r \geq 1), & p_{1,q-1} &= \omega_{q-1} / A_0, \\ p_{2,N} &= \frac{|C_r|^2}{A_r (\nu_r - \omega_{N+1})}, & \frac{p_{2,N}}{p_{1,N}} &= \frac{|C_r / C_{r-1}|^2 \nu_{r-1} (\nu_{r-1} - \omega_N)}{\omega_N (\nu_r - \omega_{N+1})} \quad (r \geq 1). \end{aligned} \quad (\text{RBE16})$$

None of the positive multipliers in the ratio is discarded. These are exact original-moment evaluations: write $\hat{\chi}(y) = i^{-q} \chi(c + iy)$ and $z = (\lambda - c)/i$. Then $\hat{\chi}(z) = 0$, and $\hat{\chi}(\bar{z}) = 0$, since the full polynomial is real in y . Both C_j and Z_λ in (RBE12) therefore have polynomial integrands after cancellation of the displayed factors $y - z$, $y - \bar{z}$ against the full $\hat{\chi}(y)^2$. Orthogonal polynomials and their norms come from the original finite moment Gram. Thus (RBE16) involves no new meromorphic integral or unspecified evaluation functional.

There is also a useful exact two-term relation, stronger than a leading-coefficient identity. The original recurrence is $(S - c)\pi_j = \pi_{j+1} - (\nu_j / \nu_{j-1})\pi_{j-1}$. Conjugation on the physical line gives $C_{j+1} = -(d + it)C_j + (\nu_j / \nu_{j-1})C_{j-1} - \nu_0 \mathbf{1}_{j=0}$, omitting the nonexistent $j = -1$ term. Substitution in (RBE13) cancels every lower coefficient, leaving, for $r \geq 1$,

$$(S - \lambda)f_N = -\frac{\chi}{H_\lambda \nu_{r-1}} \{C_{r-1}\pi_r + C_r \pi_{r-1}\}. \quad (\text{RBE17})$$

At $r = 0$ the right relation is exactly χ/H_λ . The recurrence follows by pairing $\bar{\pi}_{j+1}$ against $(S - \lambda)^{-1}$, using $\bar{S} - c = -(S - c)$ and orthogonality of π_j to the constant for $j > 0$. This proves every coefficient and the boundary constant at $j = 0$. In particular (RBE17) evaluates the full source second moment:

$$\frac{\|(S - \lambda)f_N\|_a^2}{E_N} = \frac{|C_{r-1}|^2 \nu_r + |C_r|^2 \nu_{r-1}}{A_r \nu_{r-1}^2} \quad (r \geq 1). \quad (\text{RBE18})$$

Both orthogonal relation components have been retained.

27 Exact adjacent-cutoff transport of the two energies

Use the distinct notation $\varrho_{N+1} = \det G_{N+1} / \det G_N$ for the original covariance contraction, and set $\xi_N = 1 - \varrho_{N+1}$. The matrix inverse method is credited to Hager [19, equation (2), p. 221]; the following proof derives its actual source vector and denominator. The covariance sum in the orthogonal source basis increases by $b_{N+1} b_{N+1}^* / \omega_{N+1}$. Direct multiplication verifies its inverse

update:

$$\begin{aligned} G_{N+1} &= G_N - \frac{G_N b_{N+1} b_{N+1}^* G_N}{\omega_{N+1} + F_{22}}, \quad \varrho_{N+1} = \frac{\omega_{N+1}}{\omega_{N+1} + F_{22}} = \frac{\omega_{N+1}}{\nu_r}, \\ \eta_N &= 1 - \frac{E_{N+1}}{E_N} = \xi_N p_{2,N}, \quad p_{1,N+1} = \frac{\varrho_{N+1} p_{2,N}}{1 - \xi_N p_{2,N}}. \end{aligned} \quad (\text{RBE19})$$

The determinant formula follows by completing the rank-one direction to a basis, or by multilinearity. The class-norm identity follows by pairing the first line with the same v_λ . At the next cutoff the first boundary column is exactly the old b_{N+1} ; its squared norm becomes $\varrho_{N+1} F_{22}$, and its pairing becomes $\varrho_{N+1} F_{23}$. Its retained-class norm becomes $E_N(1 - \xi_N p_{2,N})$. Substituting these three actual values proves the final formula. All contractions are in $(0, 1)$, and $0 < \eta_N < 1$, by the strictly positive next Gram and the nonzero F_{23} .

The phase now evaluates this drop exactly. The subtracted relation $\chi\pi_r$ in (RBE14) has the old g_N -parity, because $q + r = N + 1$. Therefore the entire old e_N -parity component of f_N is unchanged as a polynomial at the next cutoff. That parity is the next g_{N+1} -parity. Taking its actual squared norm gives $u_N E_N = v_{N+1} E_{N+1}$. Insert (RBE6) to obtain

$$\frac{E_{N+1}}{E_N} = \frac{\mu_N}{2t - \mu_{N+1}}, \quad \eta_N = 1 - \frac{\mu_N}{2t - \mu_{N+1}}, \quad p_{2,N} = \frac{1 - \mu_N/(2t - \mu_{N+1})}{1 - \varrho_{N+1}}. \quad (\text{RBE20})$$

These identities hold at every finite $N \geq q - 1$; both denominators are positive by (RBE7). They relate the current to the individual energy through the original minimum, rather than through a selected trial vector.

Define, at every such cutoff, the positive directed phase number

$$\varepsilon_N = \begin{cases} 2t - \mu_N, & r = N + 1 - q \text{ even,} \\ \mu_N, & r = N + 1 - q \text{ odd.} \end{cases} \quad \mathbf{c}_N = d + i(-1)^r(t - \varepsilon_N). \quad (\text{RBE21})$$

This is a definition at all cutoffs and agrees with the evaluated EC34 error on its original window. Formula (RBE20) becomes

$$\eta_N = \begin{cases} (\varepsilon_N - \varepsilon_{N+1})/(2t - \varepsilon_{N+1}), & r \text{ even,} \\ 1 - \varepsilon_N/\varepsilon_{N+1}, & r \text{ odd.} \end{cases} \quad (\text{RBE22})$$

Since $0 < \eta_N < 1$, it proves the strict alternating inequalities $\varepsilon_{N+1} < \varepsilon_N$ for even r , and $\varepsilon_N < \varepsilon_{N+1}$ for odd r . No ratio of these positive errors is assigned from separate upper bounds. Equations (RBE19)–(RBE22) determine both boundary energies from the actual consecutive phase values and the original covariance contractions. The first cutoff keeps its separate formula (RBE16).

28 Finite guards and actual degree estimates on both parities

Here are the unchanged finite EC data [18]. Put $R = a\sqrt{\delta^2 + \gamma^2}$, $Y = 2^{-16}q$, $B_\chi = qR^2/(Y^2 - R^2)$, and retain the original full-polynomial comparison logarithm $\mathcal{L}_a = \log(u_a/\ell_a)$. With the original Gamma lower constant $c_\Gamma = \sqrt{2}e^{-7/6}$, define

$$\begin{aligned} \eta_0 &= \frac{\sqrt{2\pi}}{c_\Gamma} \frac{(2q+1)(4q+1)\sqrt{1+2q}}{q} 16^{-q}, \quad \eta = e^{\mathcal{L}_a + B_\chi} \frac{\eta_0}{1 - \eta_0}, \quad g^2 = Y^2(1 - \eta), \\ C_* &= C_{\text{mult}}(h)(4q + \lfloor a/3 \rfloor), \quad t_I = \frac{R^2}{Y^2 - R^2} + \frac{\eta C_*^2}{dR(1 - \eta)^2}, \quad a_* = [(1 - R^2/g^2)(1 - t_I)^2]^{-1} - 1, \\ E_{\text{cur}} &= \max \left\{ \frac{2R^2 a_*}{d}, \quad 2t \left[\frac{R^2}{Y^2} + \frac{\eta(R^2 + C_*^2)}{d^2} \right] \right\}. \end{aligned} \quad (\text{RBE23})$$

The source guards are $q \geq 100$, $R \leq Y/4$, $\eta \leq 1/4$, $t_I < 1$, and $Ra_* \leq d/2$. On them EC proves $0 < \varepsilon_N \leq E_{\text{cur}} = O_h(a^3/q^2)$ for every $q-1 \leq N \leq 2q-1$, retaining the full root polynomial, both source regions, and all multiplicities. The current work uses that established finite estimate; no new Cauchy asymptotic is assumed in place of its original positive multipliers.

Its directed bound extends to $N = 2q$ by an exact new receiver. Indeed q is even, so the final old cutoff $N = 2q-1$ has $r = q$ even. Formula (RBE22) gives

$$0 < \varepsilon_{2q} < \varepsilon_{2q-1} \leq E_{\text{cur}}. \quad (\text{RBE24})$$

This uses the actual next finite section and does not extend a degree-limited analytic estimate by notation.

Add only the explicit eventual guard $E_{\text{cur}} \leq t/2$, and put $\zeta = E_{\text{cur}}/(2t) \leq 1/4$. For $q-1 \leq N \leq 2q-1$, (RBE7) and (RBE21) give

$$\begin{aligned} r \text{ even : } & p_{2,N} \leq v_N = \varepsilon_N/(2t) \leq \zeta, \quad 0 < \eta_N \leq \zeta, \\ r \text{ odd : } & p_{1,N} \leq u_N = \varepsilon_N/(2t) \leq \zeta. \end{aligned} \quad (\text{RBE25})$$

The odd bound can be substantially sharpened through the original arithmetic covariance. CAI25–27 gives, with the unchanged exterior threshold $L = \delta q(a+1)/2$,

$$\begin{aligned} B_{h,a} &= C_{\text{mult}}(h)^2(4q + \lfloor a/3 \rfloor - 2)^2, \quad z_{h,a} = B_{h,a}/L^2 = \frac{4C_{\text{mult}}(h)^2[4 + (\lfloor a/3 \rfloor - 2)/q]^2}{\delta^2(a+1)^2}, \\ 0 < \varrho_n &\leq z_{h,a}/(1 + z_{h,a}) \quad (q \leq n \leq 2q). \end{aligned} \quad (\text{RBE26})$$

For completeness this receiving inequality uses the exact scalar recurrence $\kappa_{q-1}^2 = b_q^{\text{rec}}(\varrho_q^{-1} - 1)$ and, for $N \geq q$, $\kappa_N^2 = b_{N+1}^{\text{rec}}(1 - \varrho_N)(\varrho_{N+1}^{-1} - 1)$, where $b_n^{\text{rec}} = \omega_n/\omega_{n-1}$. These follow directly from (RBE15) and $\kappa_N^2 = F_{11}F_{22}/\omega_N^2$. PT7 and PT23 give $\kappa_N \geq L$. Dropping only the factor $1 - \varrho_N \leq 1$ in the displayed upper expression then gives $\varrho_n \leq b_n^{\text{rec}}/(b_n^{\text{rec}} + L^2)$. The multiplication estimate bounds $b_n^{\text{rec}} \leq B_{h,a}$, proving (RBE26) on every original degree, with no replacement of the arithmetic recurrence by the Gamma recurrence.

In (RBE19), the map $(s, p) \mapsto sp/[1 - (1-s)p]$ is increasing in each variable on $0 < s < 1$, $0 < p < 1$, as its two derivatives are $p(1-p)/[1 - (1-s)p]^2$ and $s/[1 - (1-s)p]^2$. Apply it after each even cutoff and use (RBE25)–(RBE26). This proves the explicit stronger odd bound

$$p_{1,N} \leq \frac{z_{h,a}\zeta}{1 + z_{h,a} - \zeta} = O_h(q^{-2}) \quad (r = N+1-q \text{ odd}). \quad (\text{RBE27})$$

It holds at all odd cutoffs $q \leq N \leq 2q$, including the new terminal receiver (RBE24). The order follows from $z_{h,a} = O_h(a^{-2})$ and $\zeta = O_h(a^2/q^2)$, with all constants displayed above.

Both individual energies also retain the source-specific lower and upper bounds in (RBE9). On the stated phase guard, $|\mathbf{c}_N|^2 = d^2 + (t - \varepsilon_N)^2 \geq a^2(\delta^2 + \gamma^2/4)$. Since $a \leq q$ and $N \leq 2q$, $C_N \leq (13/3)C_{\text{mult}}(h)q$. Thus, at every $q-1 \leq N \leq 2q$,

$$\begin{aligned} p_{2,N} &\geq \frac{d^2 + (t - E_{\text{cur}})^2}{d^2 + (t + C_N)^2} \geq c_h^{\text{resp}} \frac{a^2}{q^2}, \quad c_h^{\text{resp}} = \frac{\delta^2 + \gamma^2/4}{\delta^2 + [\gamma + (13/3)C_{\text{mult}}(h)]^2} > 0, \\ \frac{d^2 + (t - E_{\text{cur}})^2}{\kappa_N^2} &\leq p_{1,N} \leq \frac{d^2 + (t + C_N)^2}{\kappa_N^2}. \end{aligned} \quad (\text{RBE28})$$

The inequalities through $N = 2q$ use (RBE24), not an omitted endpoint. Equations (RBE25) and (RBE28) prove the actual two-sided order $p_{2,N} = \Theta_h(a^2/q^2)$ at every even r in the original window. At odd r , (RBE27) evaluates a smaller upper scale for p_1 ; (RBE20)–(RBE22) retain the actual phase-error ratio determining p_2 . No lower order for that ratio is guessed from the separate EC error bounds.

29 Return to the original response ellipse

Let $P_K = I(I^*G_N I)^{-1}I^*G_N$ be the original observation- kernel projection, with its full period-dependent inclusion I . Keep $\theta_N = \|P_K v_\lambda\|_{G_N}^2 / E_N$ and the actual ratios $U_N = K_{13}/F_{13}$, $V_N = K_{23}/F_{23}$. The original JRE21 ellipse factors, denoted here by $\mathfrak{d}_N, \mathfrak{h}_N$ to avoid collision with the eigenvalue displacement, receive the new energies exactly:

$$\mathfrak{d}_N^2 = \theta_N(1 - \theta_N)(p_{1,N}^{-1} + p_{2,N}^{-1}), \quad \mathfrak{h}_N = \frac{\theta_N(1 - \theta_N)\kappa_N}{2|\mathfrak{c}_N|} \sqrt{1 - p_{1,N} - p_{2,N}}. \quad (\text{RBE29})$$

The second formula substitutes the exact existing product in (RBE7), rather than treating the two energies independently. In particular the actual minimum variance from (RBE8) supplies p_2 , (RBE16) supplies both Cauchy factors, and (RBE19)–(RBE22) supply their adjacent-cutoff transport. The amplification in JRE22 still contains the actual θ_N, U_N, V_N , and κ_N ; none is set to a rank fraction or to a total trace. Its original three signed errors still sum to zero.

To make the index map to the next response explicit, the first column at $N+1$ is literally b_{N+1} , the old second column, and (RBE19) gives $F'_{13} = \varrho_{N+1}F_{23}$. Thus an exact update of the original kernel residual pairing $B'_{13} = (\varrho_{N+1}/d_K)B_{23}$, with its actual kernel determinant ratio d_K , yields exactly $U_{N+1} = 1 - (1 - V_N)/d_K$. This is the receiving substitution into the separately derived original-kernel update; no conjugation or exchange of boundary rows is required.

Metric condition cost and the exact marked observation

This is the complete receiving proof for Sections 4–5 of the supplied low-cutoff continuation [22]. LCS1–40, ORE1–23 and the independent conductor calculation supply its original source objects and estimates. PT1–41 [23] supplies the full-primary positive-trace minimum, including all nilpotent blocks. The final selected-minor conclusion is corrected through the exact COI11 map [24] instead of silently identifying two coefficient sections. All observations below retain their original period.

30 The cost of a positive-trace metric

Use the full original quartet polynomial with fixed multiplicity, $e = 1 + k(m_0 - 1)$, $q = e(k + 1)^2$, $c = k/2$, and source cutoff $q - 1$. Let $G_0 = G_{q-1}$ denote its original arithmetic minimum metric. In its orthonormal coordinates the full multiplication operator satisfies the proved original boundary identity NCE10:

$$\begin{aligned} B_0 &= G_0^{1/2}(A - cI)G_0^{-1/2} = iJ + \epsilon_0 f e^*, \\ J &= J^*, \quad \|e\| = \|f\| = 1, \quad e^* f = 0, \quad \epsilon_0^2 = \frac{\nu_0 - \omega_q}{\omega_{q-1}}. \end{aligned} \tag{CMO1}$$

In this section the two occurrences of e have different types: the integer multiplicity in the first paragraph and the unit boundary vector in CMO1. The integer is never used as a vector. The exact source Gram and LCS9 give

$$\epsilon_0^2 = \frac{1 - d_0}{m_z p_{1,q-1}}, \quad \epsilon_0 = \frac{4q}{\pi \sqrt{p_{1,q-1}}} (1 + o_h(1)). \tag{CMO2}$$

All norms retain their source mass. The asymptotic follows from $Y^2 m_z \rightarrow 1$, $Y = 4q/\pi$, and $d_0 \rightarrow 0$, proved in LCS19,30. The original multiplication bound gives

$$\|J\| \leq J_* = C_{\text{mult}}(h)(2q - 2 + \lfloor k/3 \rfloor).$$

For any positive Hermitian metric $\widehat{G}$ on the same original coefficient space set

$$M = G_0^{-1/2} \widehat{G} G_0^{-1/2}, \quad S = M^{1/2}, \quad \kappa = \text{cond } M, \quad B_S = S B_0 S^{-1}, \quad H_S = B_S + B_S^*.$$

Thus S is the exact isometry from the new metric in the old orthonormal coordinates to a Euclidean coordinate space. The weight has no extra factor $1/2$. Let $\mathcal{P}(\widehat{G}) = \text{Tr}(H_S)_+$. The spectral trace is invariant under this coordinate identification, so it equals the positive trace for $\widehat{G}$. The original symmetric root multiset has $\Re \text{Tr } B_0 = 0$, including every repeated primary; hence $\text{Tr } H_S = 0$. If its real eigenvalues are h_j , this says $\sum_{h_j > 0} h_j = -\sum_{h_j < 0} h_j$. Every $|h_j|$ is bounded by that common sum, proving $\|H_S\| \leq \mathcal{P}(\widehat{G})$.

Put $u = Sf$, $w = S^{-*}e$. Then $w^*u = e^*f = 0$ and

$$S B_0 S^{-1} = i S J S^{-1} + \epsilon_0 u w^*.$$

On the orthonormal pair $u/\|u\|$, $w/\|w\|$ the Hermitian rank-two matrix $uw^* + wu^*$ has eigenvalues $\pm \|u\| \|w\|$; it is zero on their orthogonal complement. Moreover $\|u\| \|w\| \geq \sigma_{\min}(S)/\sigma_{\max}(S) = \kappa^{-1/2}$, while $\|i S J S^{-1} + (i S J S^{-1})^*\| \leq 2J_* \sqrt{\kappa}$. The triangle inequality applied to this exact decomposition proves

$$\boxed{\epsilon_0 \leq \mathcal{P}(\widehat{G}) \sqrt{\kappa} + 2J_* \kappa.} \tag{CMO3}$$

The classical Hermitian norm and quadratic-coordinate framework is credited to Boyd and Vandenberghe [21, Appendix A.5]; the entire rank-two comparison just used is explicitly proved here.

PT1–41 evaluates the full original minimum as $L_{h,k} = \delta q(k+1)/2$. For every attaining metric, put $x = \sqrt{\kappa}$. CMO3 gives $2J_*x^2 + L_{h,k}x - \epsilon_0 \geq 0$. Since $x > 0$ and $J_* > 0$, solving the quadratic with its positive root and rationalizing proves the finite bound

$$\kappa \geq \left(\frac{2\epsilon_0}{L_{h,k} + \sqrt{L_{h,k}^2 + 8J_*\epsilon_0}} \right)^2. \quad (\text{CMO4})$$

LCS19 gives $\log \epsilon_0 = q \log(4/\pi) + O_h(k + \log q)$. Thus $L_{h,k}^2/(J_*\epsilon_0) \rightarrow 0$ at every fixed multiplicity, since $L_{h,k}$ and J_* are polynomial in k . CMO2 and $J_*/q \rightarrow 2C_{\text{mult}}(h)$ now give

$$\kappa \geq \frac{1 - o_h(1)}{\pi C_{\text{mult}}(h) \sqrt{p_{1,q-1}}}, \quad \liminf_{k \rightarrow \infty} \frac{\log \kappa}{q} \geq \log(4/\pi) > 0. \quad (\text{CMO5})$$

The same conclusion follows for any family with $\mathcal{P}(\widehat{G})$ bounded by a polynomial in q : replace $L_{h,k}$ by that actual value in the positive root. This does not change the full multiplication operator or remove nilpotents. It quantifies the isometry’s distortion in its original metric, while preserving the valid positive-trace minimum.

31 Conductor detection on original low degrees

Now retain the simple-packet conductor domain of FEX1–4: $q = (k+1)^2$, $q' = (k-7)^2$, original $\Lambda_k : E \rightarrow B = \text{im } \Lambda_k$, $K = \ker \Lambda_k$, conductor

$$\mathcal{T}_A P(S') = \sum_{\nu} a_{\nu} P(S' + b_{\nu}), \quad \mu_j = \sum_{\nu} a_{\nu} b_{\nu}^j, \quad v = \min\{j : \mu_j \neq 0\} \leq 80.$$

Here v is the conductor order. Every representative of a class in K satisfies $\chi_{k-8} \mid \mathcal{T}_A P$. This implication descends to E : adding $\chi_k H$ vanishes at each shifted lower root, exactly as proved in the preceding conductor source. Define the literal conductor map $C : E \rightarrow \mathbb{C}[S']/(\chi_{k-8})$ by $C[P] = [\mathcal{T}_A P]$. Then $K \subset \ker C$, so there is a unique linear map

$$\bar{C} : B \longrightarrow \mathbb{C}[S']/(\chi_{k-8}), \quad \bar{C}(\Lambda_k x) = Cx, \quad C = \bar{C}\Lambda_k. \quad (\text{CMO6})$$

Well-definedness follows because any two lifts differ by K ; surjectivity of Λ_k onto B proves uniqueness. This is an exact typed factorization of the original maps.

If P has degree $D \geq v$ and nonzero leading coefficient, Taylor expansion gives

$$\deg \mathcal{T}_A P = D - v, \quad \text{lc}(\mathcal{T}_A P) = \mu_v \binom{D}{v} \text{lc}(P) \neq 0.$$

For $v \leq D < q' + v$ this degree is below q' , so divisibility by χ_{k-8} is impossible. If $D < v$, every derivative term carrying a nonzero moment vanishes. Consequently, with $\mathcal{P}_{-1} = 0$,

$$K \cap \mathcal{P}_D = K \cap \mathcal{P}_{v-1} \quad (v-1 \leq D < q' + v), \quad CS^v = \mu_v[1] \neq 0. \quad (\text{CMO7})$$

In particular

$$j_* = \min\{j : \Lambda_k S^j \neq 0\} \leq v \leq 80. \quad (\text{CMO8})$$

All polynomial spaces here are included by their actual classes in E , so there is no quotient-metric identification in CMO7.

32 The marked derivative and its exact observed coefficient

Use the marked family in the original increasing-power frame:

$$A(t) = A + te_0 e_{q-1}^\top, \quad U'(t) = A(t)U(t)/u, \quad U(0) = I, \quad u \neq 0, \quad \Lambda_h(t) = \Lambda_k U(t)^{-1}.$$

Writing $E = e_0 e_{q-1}^\top$, the coefficients of $U(t) = \sum_{n \geq 0} U_n t^n$ satisfy $U_0 = I$, $U_{-1} = 0$ and $(n+1)U_{n+1} = (AU_n + EU_{n-1})/u$. Their norms are termwise bounded by the nonnegative coefficients of $\exp((\|A\|t/|u| + (\|E\|t^2/(2|u|)))$, since those coefficients satisfy the corresponding scalar recurrence. Thus the series converges for every t , solves the equation term by term, and is unique by the same recurrence. It has an analytic inverse near 0, since its determinant starts at one. Let $v_\lambda = [b_\lambda \chi_k / (S - \lambda)]$ be the full terminal eigenclass and put $Y(t) = e^{\lambda t/u} \Lambda_h(t) v_\lambda - \Lambda_k v_\lambda$. Differentiating the inverse on its actual domain gives $\Lambda'_h = -\Lambda_h A(t)/u$. Since $Av_\lambda = \lambda v_\lambda$ and $e_{q-1}^\top v_\lambda = b_\lambda$, we obtain

$$Y'(t) = -\frac{b_\lambda}{u} t e^{\lambda t/u} \Lambda_h(t) 1. \quad (\text{CMO9})$$

On the admitted conductor domain $k \geq 9$, we have $q \geq 100 > v+1$ since $v \leq 80$. For $j \leq v$, the Taylor recurrence from $\Lambda'_h = -\Lambda_h(A + te_0 e_{q-1}^\top)/u$ yields

$$\Lambda_h^{(j)}(0)1 = (-u)^{-j} \Lambda_k S^j. \quad (\text{CMO10})$$

To justify the omitted marked terms explicitly, each term containing $E = e_0 e_{q-1}^\top$ and acting on $1 = e_0$ has a rightmost E preceded on its right by fewer than $q-1$ copies of A . But $A^r 1 = S^r$ for $r < q$ and $e_{q-1}^\top S^r = 0$ for $r < q-1$. Every such term vanishes. The remaining term is $(-u)^{-j} \Lambda_k A^j 1$, proving CMO10 with its sign.

For $j < j_*$ all observed Taylor coefficients vanish. Leibniz's rule at j_* in CMO9 therefore gives the first nonzero derivative

$$\boxed{Y^{(j_*+2)}(0) = b_\lambda(j_*+1)(-u)^{-(j_*+1)} \Lambda_k S^{j_*} \neq 0, \quad j_*+2 \leq v+2 \leq 82.} \quad (\text{CMO11})$$

Applying $\bar{C}$, the same reasoning uses $CS^j = 0$ for $j < v$ and $CS^v = \mu_v[1]$. It proves the explicit coefficient

$$\bar{C}Y^{(r+2)}(0) = 0 \quad (r < v), \quad \boxed{\bar{C}Y^{(v+2)}(0) = b_\lambda(v+1)(-u)^{-(v+1)} \mu_v[1] \neq 0.} \quad (\text{CMO12})$$

For $v = 0$ this also proves $\Lambda_k 1 \neq 0$ on every admitted conductor degree. No arithmetic projected sign is inferred from this marked analytic Taylor coefficient.

33 The precise return to the original coefficient image

COI proves $\text{im } L_0 = \mathcal{P}_{m-1}$ for its consecutive comparison section, $m = 16k - 48 - v$. At $k \geq 29$ and $0 \leq v \leq 80$, $v-1 \leq m-1 < q' + v$: the upper difference is $q' + v - (m-1) = k^2 - 30k + 98 + 2v$, positive at 29 and increasing thereafter. CMO7 therefore proves

$$K \cap \text{im } L_0 = K \cap \mathcal{P}_{v-1}, \quad \text{rank}(\Lambda_k L_0) = m - t_0, \quad t_0 = \dim(K \cap \mathcal{P}_{v-1}). \quad (\text{CMO13})$$

For the original selected minor I , use the proved map and its observed image, without dropping the conductor term:

$$\boxed{L_I = F_C D_{0I} + L_0 M_{0I}, \quad \Lambda_k L_I = (\Lambda_k F_C) D_{0I} + (\Lambda_k L_0) M_{0I}.} \quad (\text{CMO14})$$

Here M_{0I} is invertible by the full COI9–11 transition. Thus the supplied note's unqualified consecutive-image statement is replaced by its exact original-minor receiver.

Finally RBV's explicit polynomial [25] is

$$t'_0 = \frac{\pi}{2\gamma}, \quad P_k(S) = \sum_{j=0}^{4k} \frac{(t'_0)^{2j} (S-c)^{2j}}{(2j)!}.$$

It has degree $8k$, nonzero leading coefficient $(t'_0)^{8k}/(8k)!$, and the negative reflected trace proved in the cited complete source. For $k \geq 29$, $v \leq 8k < q' + v$ and $8k \leq m - 1$, directly from the above degree formulas. Its conductor has nonzero leading coefficient

$$\mu_v \binom{8k}{v} \frac{t'_0{}^{8k}}{(8k)!} \neq 0. \tag{CMO15}$$

Its degree is $8k - v < q'$, so $CP_k \neq 0$ and hence $\Lambda_k P_k \neq 0$ by CMO6. The negative-trace test survives the actual observation on this domain. Its signed residue pairing and the arithmetic projected current retain their respective exact definitions; CMO6 and CMO14 are the proved observation maps, and do not assert equality of those pairings.

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